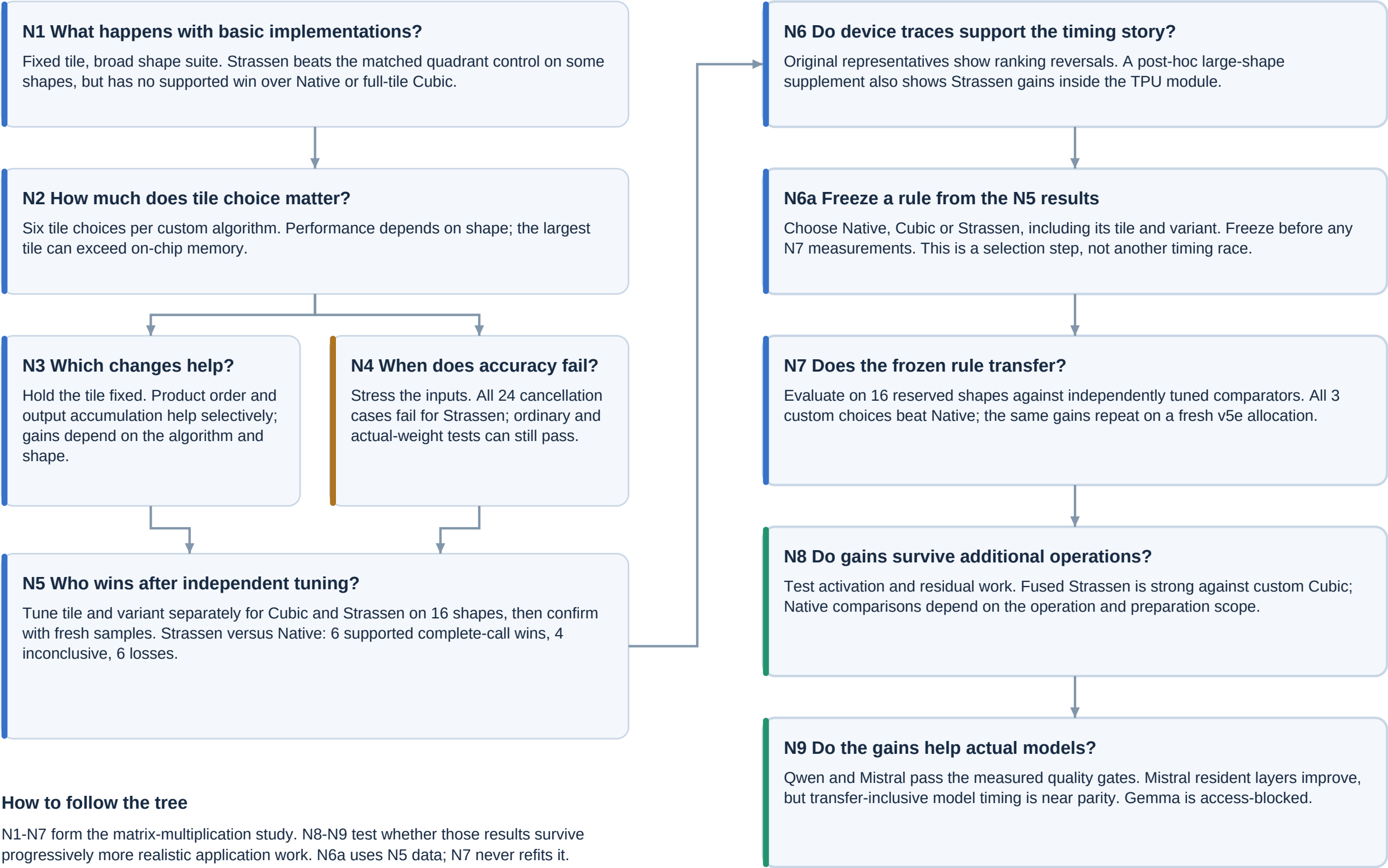


The study, question by question

Completed v5e evidence. Arrows show the research logic, not the exact launch order.



How to follow the tree

N1-N7 form the matrix-multiplication study. N8-N9 test whether those results survive progressively more realistic application work. N6a uses N5 data; N7 never refits it.

Future work: v6e replication and optional two-level Strassen. Neither is included in these results.

How to read the graphs

Start with the question, then read the bar heights, then the interpretation beneath the graph.

1. Compare within one shape

Each cluster is one matrix shape, written $M \times K \times N$ for $(M \times K)$ times $(K \times N)$. Blue is Native, orange is Cubic and green is Strassen. Extra controls and N3 optimization variants use the legend shown on their graph.

2. Find the lowest measured bar

For timing graphs, lower is faster. A short black horizontal segment marks the lowest eligible mean for that shape; a diamond identifies the bar. Different shapes get separate segments. These marks show the observed ordering, not statistical certainty.

3. Read uncertainty separately

Ordinary timing whiskers show sample standard deviation, not confidence intervals. The explanations classify supported comparisons using the archived paired 95% intervals, without multiple-comparison correction. Inconclusive does not mean equal.

4. Keep scopes and accuracy separate

Complete call includes required device preparation and finishing. Prepared kernel reuses prepared buffers. Neither is full inference. Device traces, resident layers and streamed forwards are separate scopes. In N4 the black marker means lowest median error; red dashed lines are tolerance gates.

A marker is not a significance test

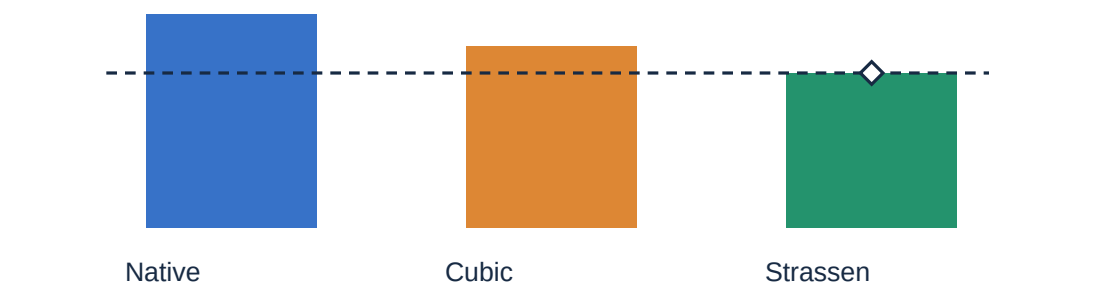


Illustration only; these are not experiment measurements.

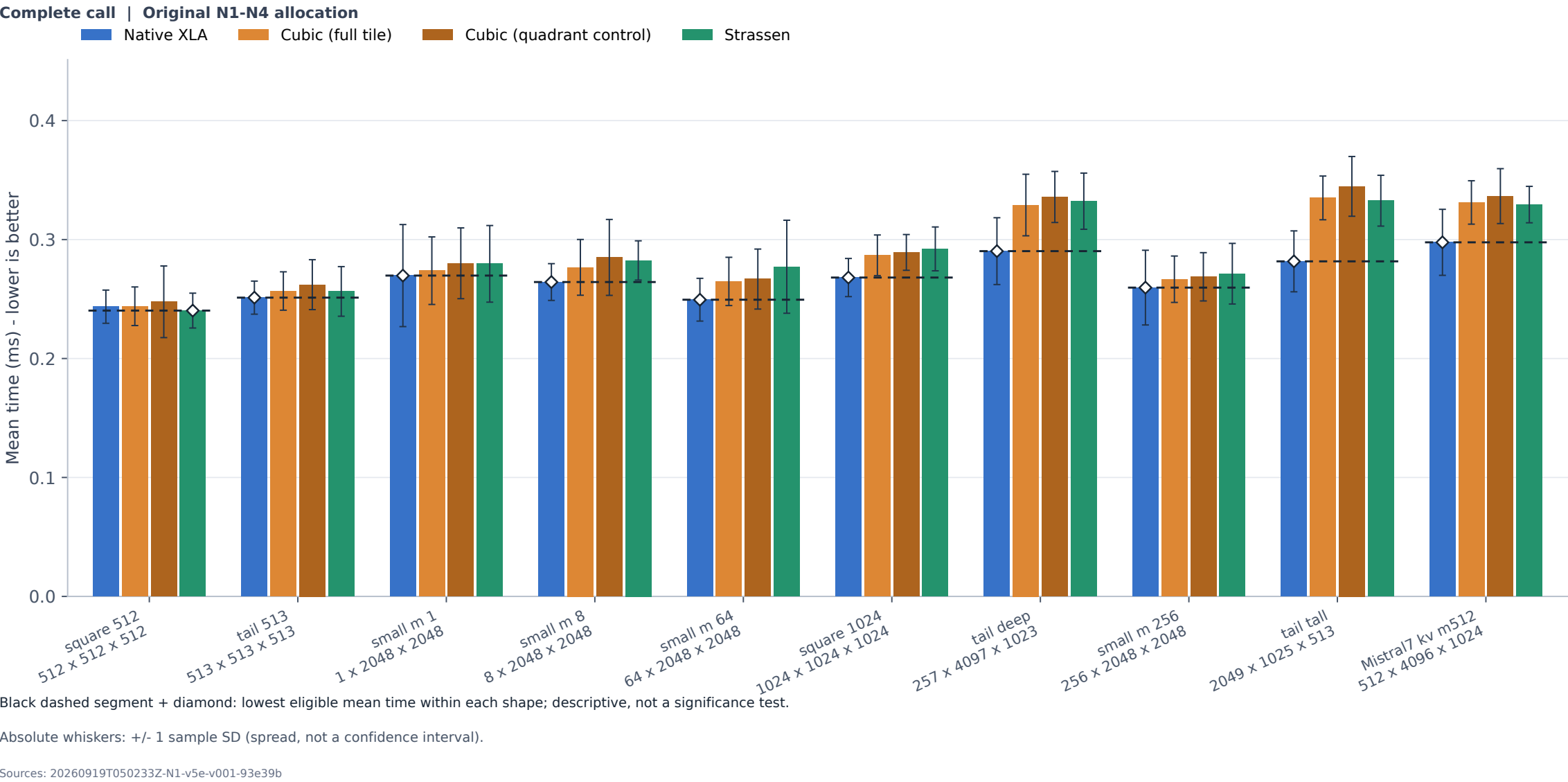
Experiment index

N1 Basic baselines	3
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Failures are retained. Selection screens are exploratory. Different allocations are never pooled. All 47 original overview comparisons are included.

N1 | Fixed-tile baselines - shapes 1-10

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.



WHAT THIS GRAPH SHOWS

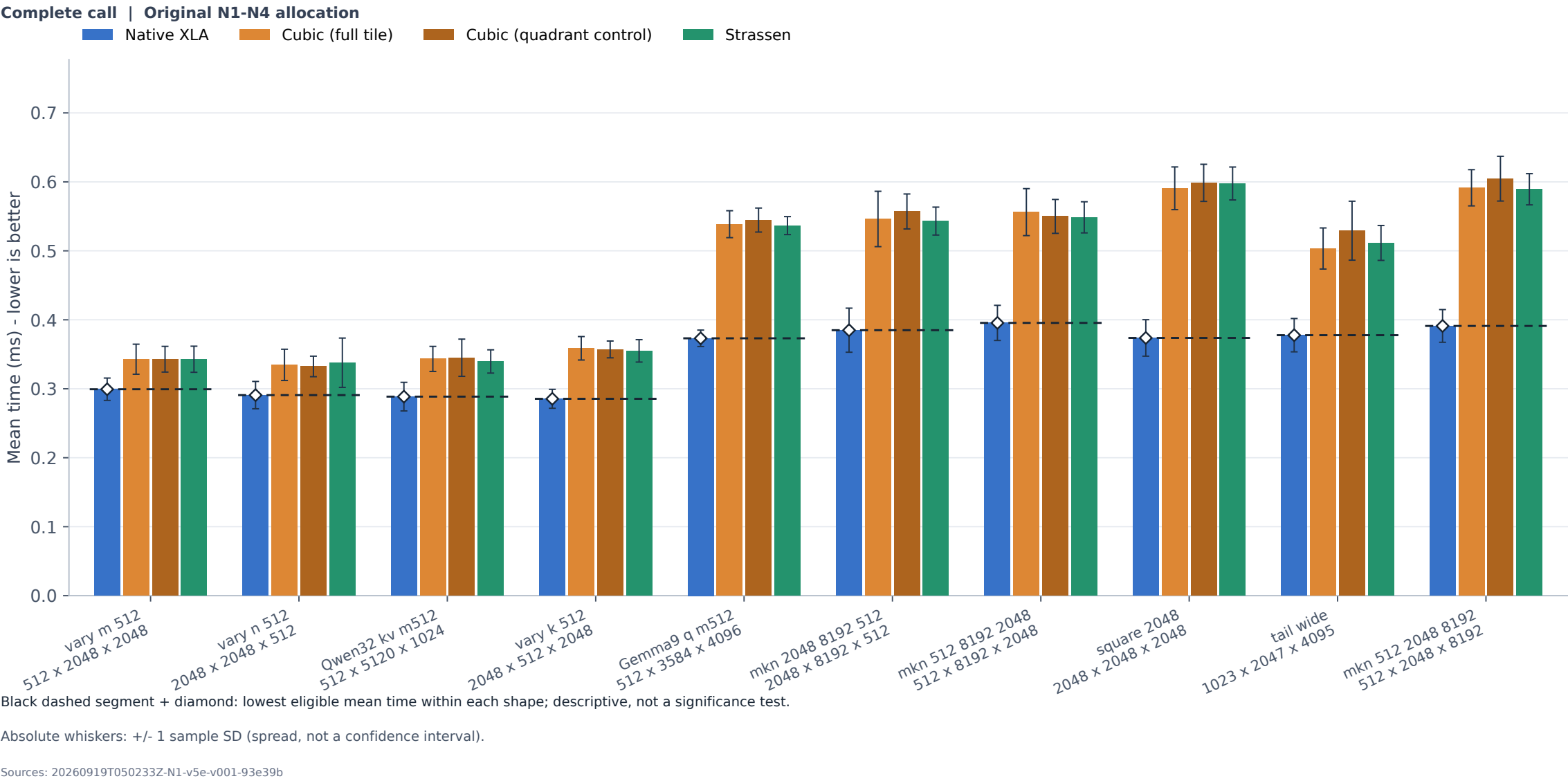
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen 1, Native XLA 9. Archived paired intervals for Strassen versus Native: 0 faster, 7 slower, 3 inconclusive.

OUR INTERPRETATION

Saving one block multiplication helps against the matched quadrant-cubic control, but is insufficient against stronger baselines. Across all 44 complete-call shapes, Strassen has 27 supported wins over the quadrant control and none over Native or full-tile Cubic. All custom kernels here use the same small, untuned 256/256/256 tile; this is a starting point, not each algorithm at its best.

N1 | Fixed-tile baselines - shapes 11-20

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.



WHAT THIS GRAPH SHOWS

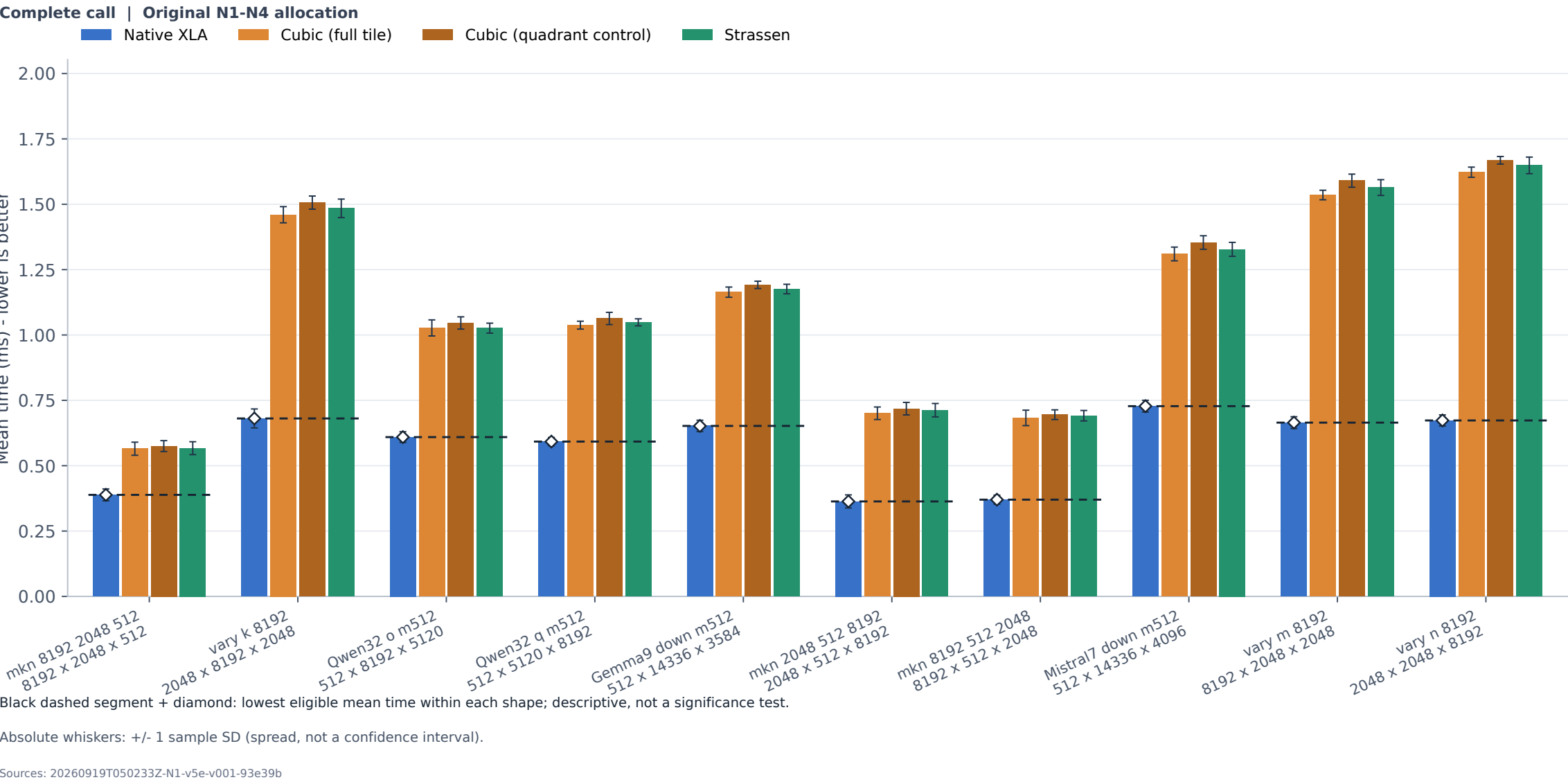
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 10. Archived paired intervals for Strassen versus Native: 0 faster, 10 slower, 0 inconclusive.

OUR INTERPRETATION

Saving one block multiplication helps against the matched quadrant-cubic control, but is insufficient against stronger baselines. Across all 44 complete-call shapes, Strassen has 27 supported wins over the quadrant control and none over Native or full-tile Cubic. All custom kernels here use the same small, untuned 256/256/256 tile; this is a starting point, not each algorithm at its best.

N1 | Fixed-tile baselines - shapes 21-30

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.



WHAT THIS GRAPH SHOWS

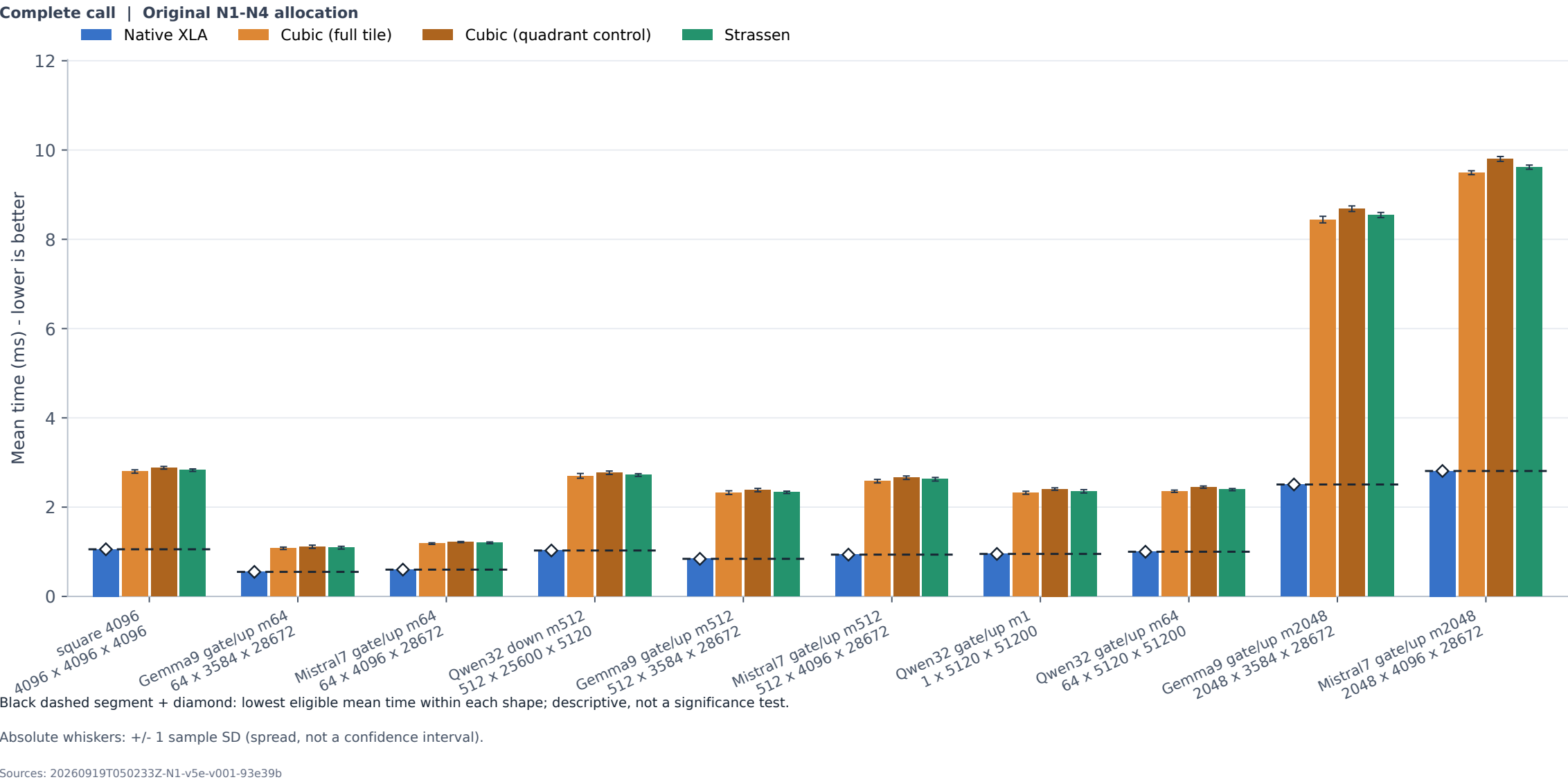
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 10. Archived paired intervals for Strassen versus Native: 0 faster, 10 slower, 0 inconclusive.

OUR INTERPRETATION

Saving one block multiplication helps against the matched quadrant-cubic control, but is insufficient against stronger baselines. Across all 44 complete-call shapes, Strassen has 27 supported wins over the quadrant control and none over Native or full-tile Cubic. All custom kernels here use the same small, untuned 256/256/256 tile; this is a starting point, not each algorithm at its best.

N1 | Fixed-tile baselines - shapes 31-40

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.



WHAT THIS GRAPH SHOWS

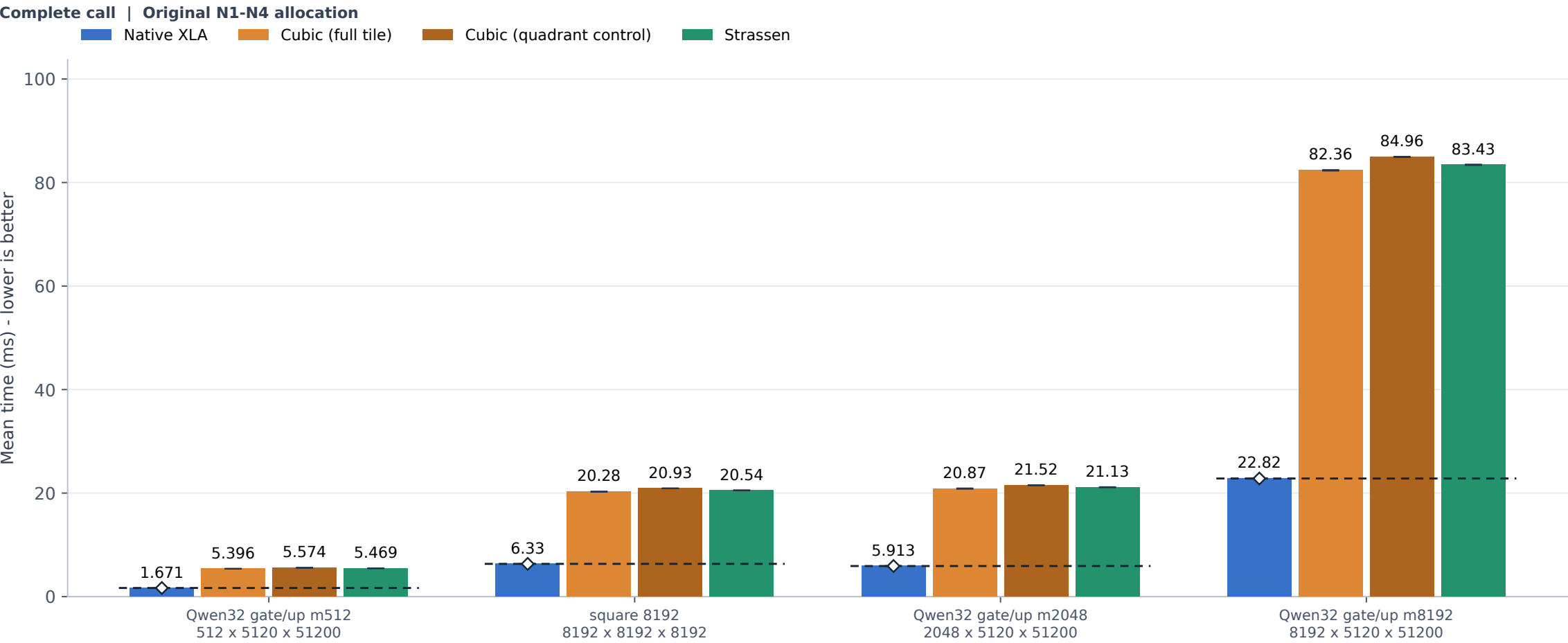
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 10. Archived paired intervals for Strassen versus Native: 0 faster, 10 slower, 0 inconclusive.

OUR INTERPRETATION

Saving one block multiplication helps against the matched quadrant-cubic control, but is insufficient against stronger baselines. Across all 44 complete-call shapes, Strassen has 27 supported wins over the quadrant control and none over Native or full-tile Cubic. All custom kernels here use the same small, untuned 256/256/256 tile; this is a starting point, not each algorithm at its best.

N1 | Fixed-tile baselines - shapes 41-44

All custom kernels use tile (BM,BN,BK)=(256,256,256). All 44 shapes are included.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T050233Z-N1-v5e-v001-93e39b

WHAT THIS GRAPH SHOWS

The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 4. Archived paired intervals for Strassen versus Native: 0 faster, 4 slower, 0 inconclusive.

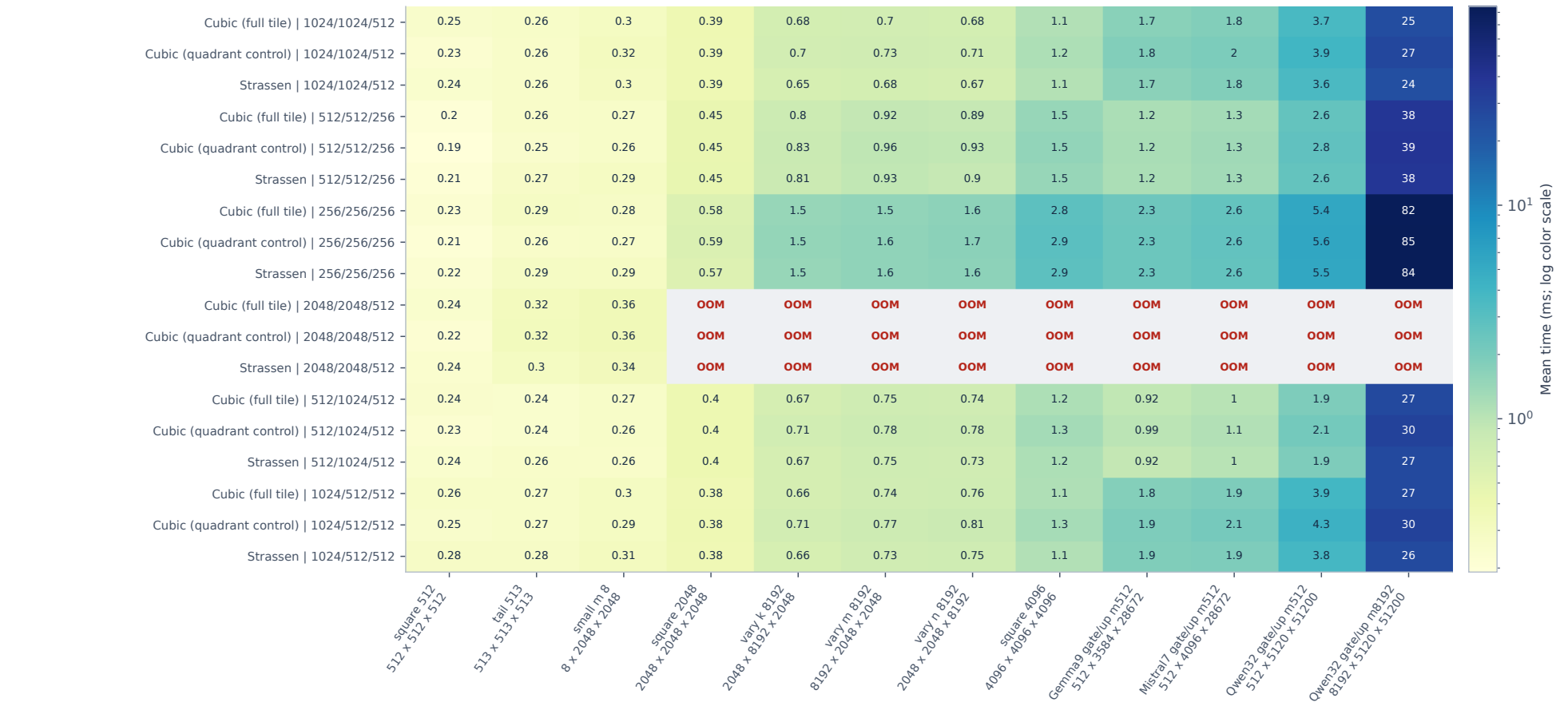
OUR INTERPRETATION

Saving one block multiplication helps against the matched quadrant-cubic control, but is insufficient against stronger baselines. Across all 44 complete-call shapes, Strassen has 27 supported wins over the quadrant control and none over Native or full-tile Cubic. All custom kernels here use the same small, untuned 256/256/256 tile; this is a starting point, not each algorithm at its best.

N2 | Every tile and algorithm

Exploratory screen: 7 timings per passing cell. Tile order is BM/BN/BK. Native was not measured.

Complete call | Original N1-N4 allocation



Numbers are archived measurements; missing cells are not measured or failed.

Sources: 20260919T050718Z-N2-v5e-v002-226e83

WHAT THIS GRAPH SHOWS

Each column is a matrix shape; each row is an algorithm/tile combination. This scope has 189 measured cells and 27 memory failures. Smaller numbers mean less time; the color scale is logarithmic. Compare algorithms within the same column, then compare tile choices. OOM means no valid timing, never zero.

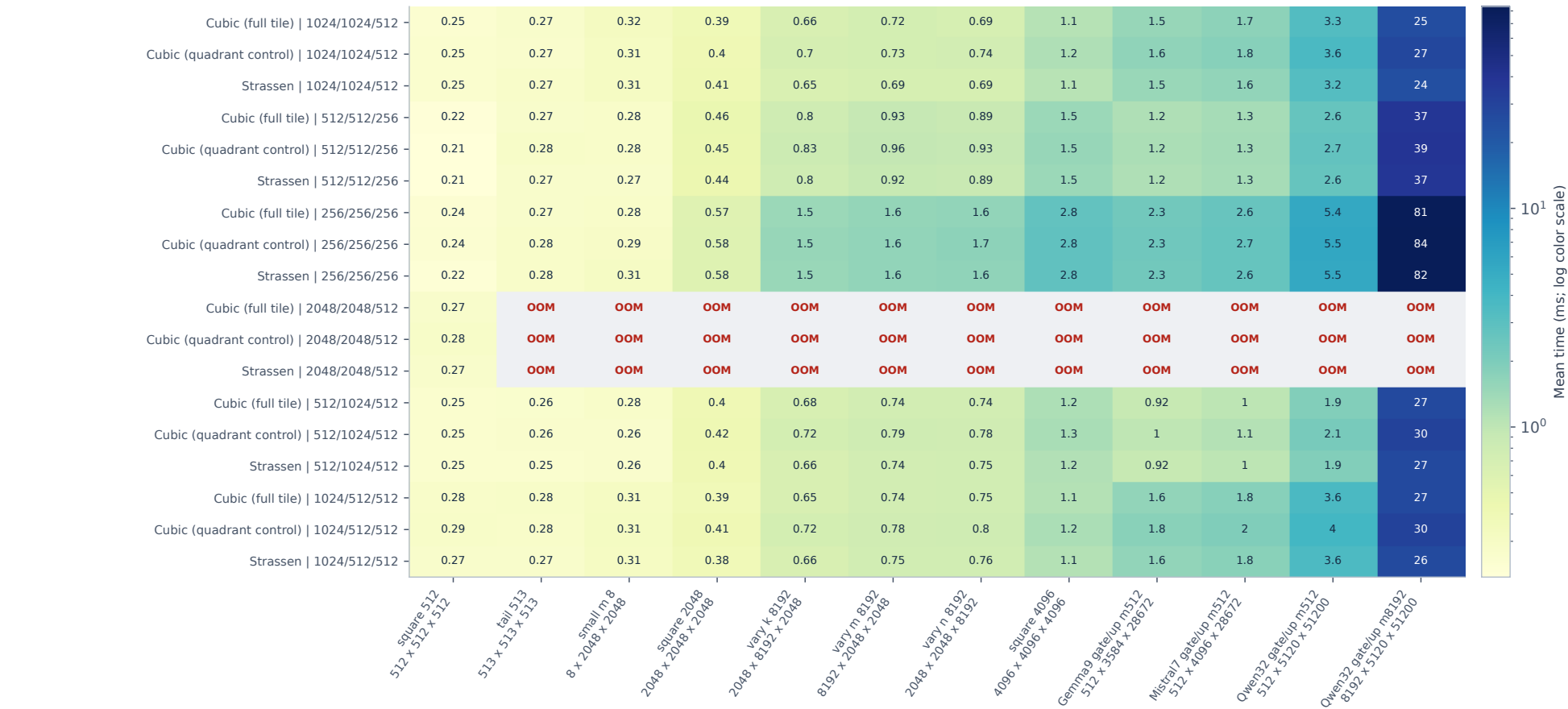
OUR INTERPRETATION

Tile size is an important performance choice, and the useful choice depends on the matrix shape and algorithm. The largest tile can exceed fast on-chip memory. These seven-repeat screening results identify candidates; they do not prove a globally optimal tile or an independently confirmed advantage. Native XLA was not measured in N2.

N2 | Every tile and algorithm

Exploratory screen: 7 timings per passing cell. Tile order is BM/BN/BK. Native was not measured.

Prepared kernel | Original N1-N4 allocation



Numbers are archived measurements; missing cells are not measured or failed.

Sources: 20260919T050718Z-N2-v5e-v002-226e83

WHAT THIS GRAPH SHOWS

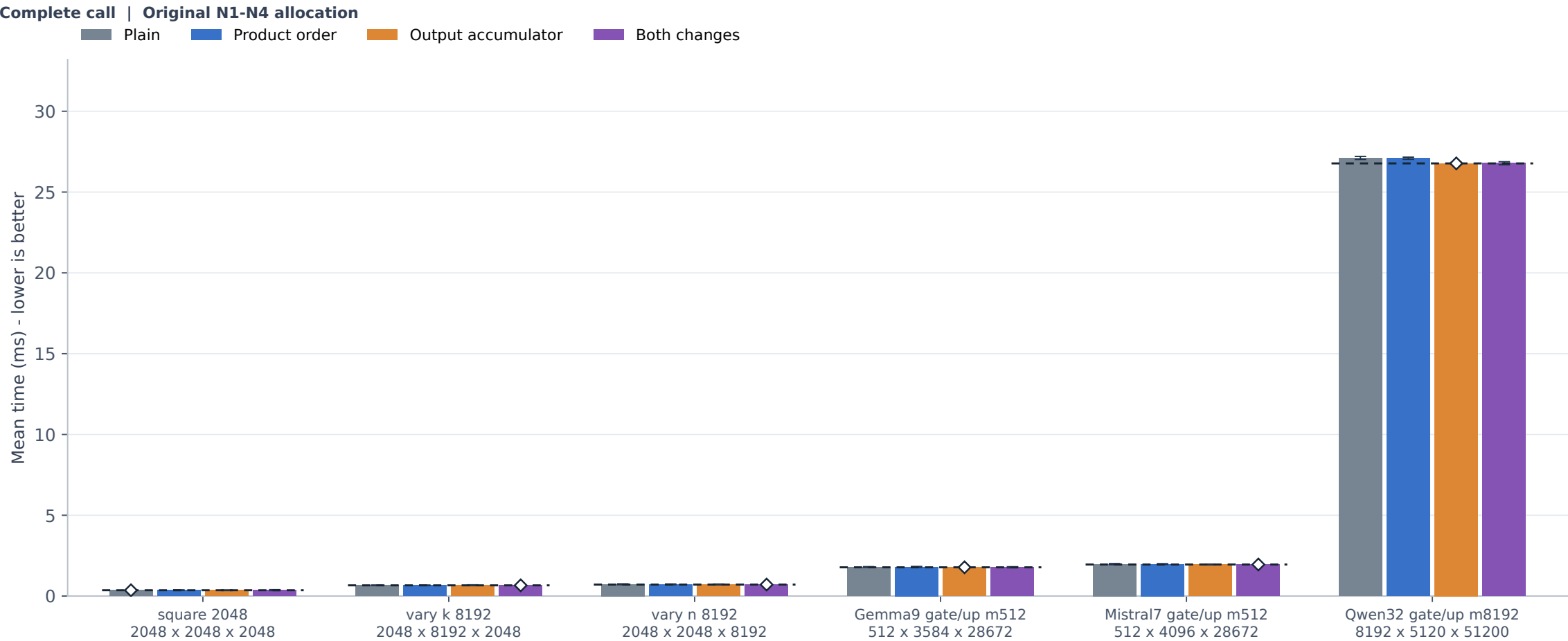
Each column is a matrix shape; each row is an algorithm/tile combination. This scope has 183 measured cells and 33 memory failures. Smaller numbers mean less time; the color scale is logarithmic. Compare algorithms within the same column, then compare tile choices. OOM means no valid timing, never zero.

OUR INTERPRETATION

Tile size is an important performance choice, and the useful choice depends on the matrix shape and algorithm. The largest tile can exceed fast on-chip memory. These seven-repeat screening results identify candidates; they do not prove a globally optimal tile or an independently confirmed advantage. Native XLA was not measured in N2.

N3 | Cubic (quadrant control) optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T051615Z-N3-v5e-v002-bc2214

WHAT THIS GRAPH SHOWS

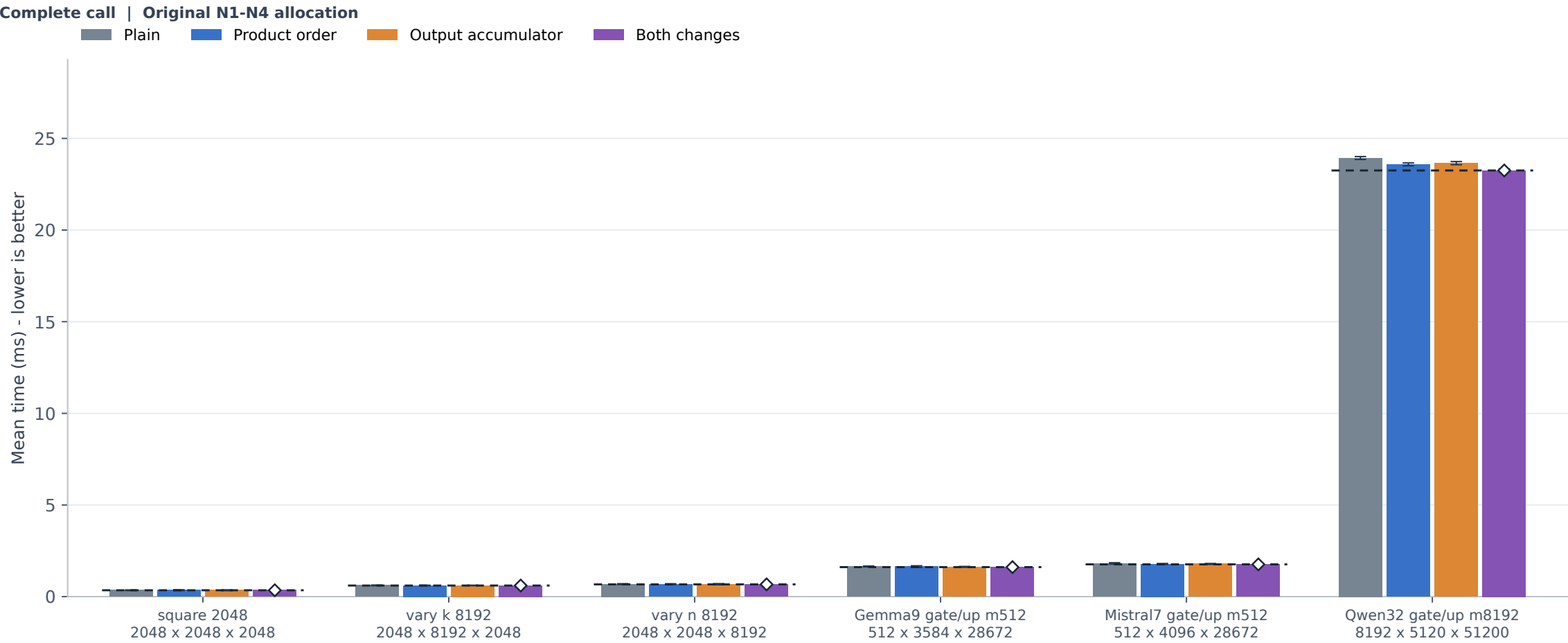
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Plain 1, Both changes 3, Output accumulator 2. A lowest mean alone is not proof of a clear win.

OUR INTERPRETATION

These changes can improve the implementation, but the effect depends on shape and algorithm. Compare each variant with its own plain version. The fixed tile is 1024/1024/512; there is no Native XLA or full-tile Cubic baseline here. Product reordering alone does not prove that the TPU overlaps vector work and matrix multiplication.

N3 | Strassen optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T051615Z-N3-v5e-v002-bc2214

WHAT THIS GRAPH SHOWS

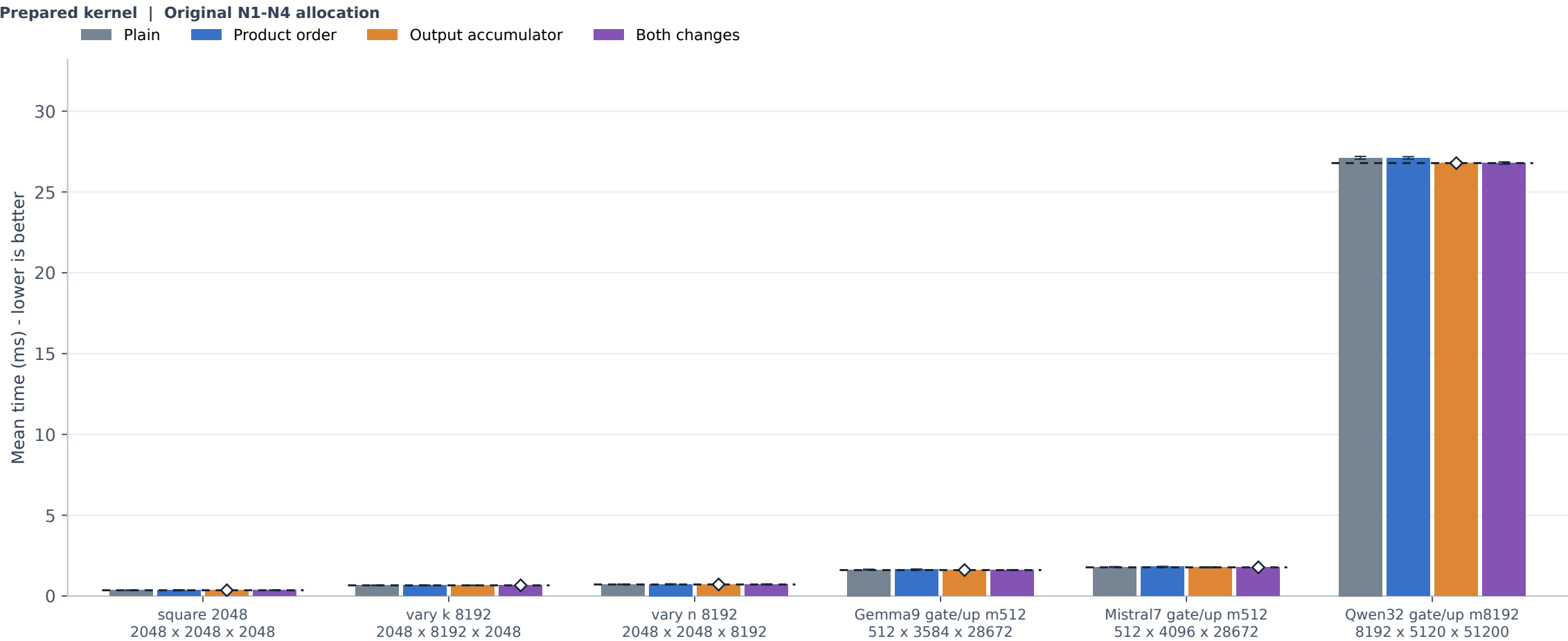
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Both changes 6. A lowest mean alone is not proof of a clear win.

OUR INTERPRETATION

These changes can improve the implementation, but the effect depends on shape and algorithm. Compare each variant with its own plain version. The fixed tile is 1024/1024/512; there is no Native XLA or full-tile Cubic baseline here. Product reordering alone does not prove that the TPU overlaps vector work and matrix multiplication.

N3 | Cubic (quadrant control) optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T051615Z-N3-v5e-v002-bc2214

WHAT THIS GRAPH SHOWS

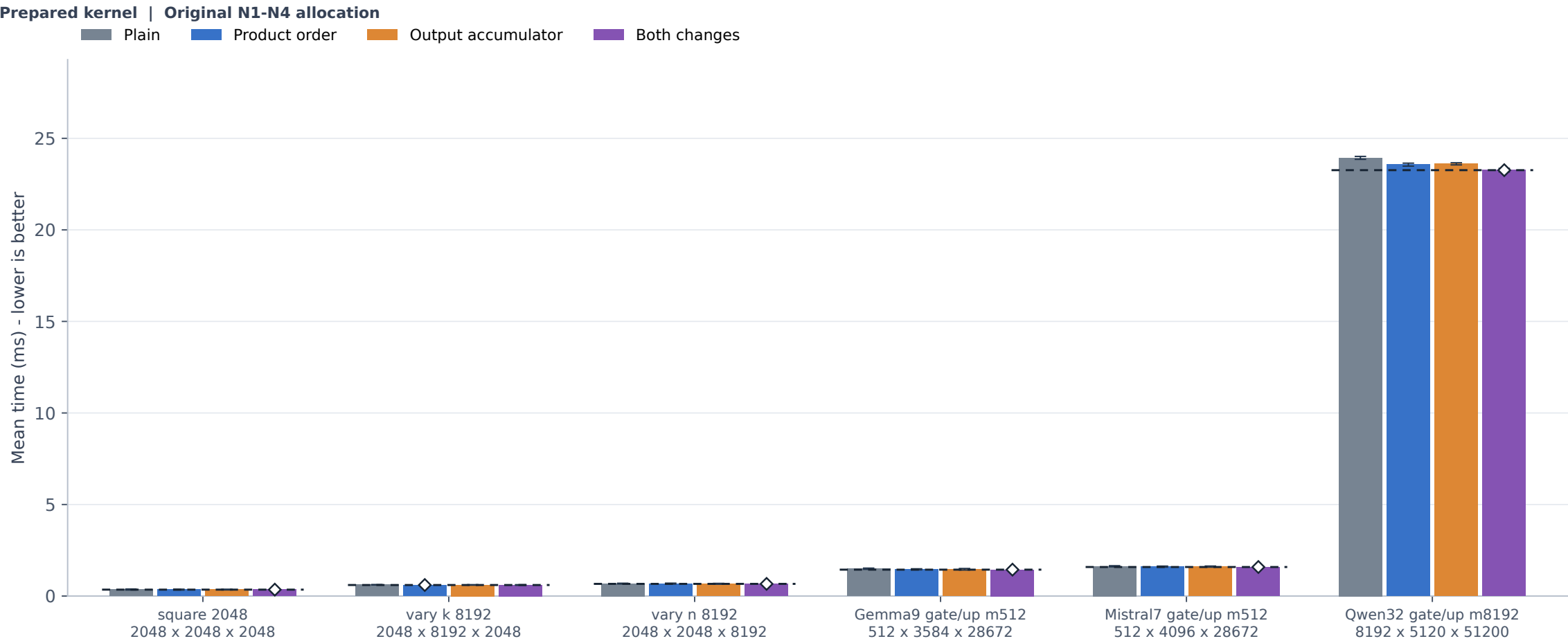
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Output accumulator 4, Both changes 2. A lowest mean alone is not proof of a clear win.

OUR INTERPRETATION

These changes can improve the implementation, but the effect depends on shape and algorithm. Compare each variant with its own plain version. The fixed tile is 1024/1024/512; there is no Native XLA or full-tile Cubic baseline here. Product reordering alone does not prove that the TPU overlaps vector work and matrix multiplication.

N3 | Strassen optimization ablation

Fixed tile 1024/1024/512. Native and full-tile cubic were not measured.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T051615Z-N3-v5e-v002-bc2214

WHAT THIS GRAPH SHOWS

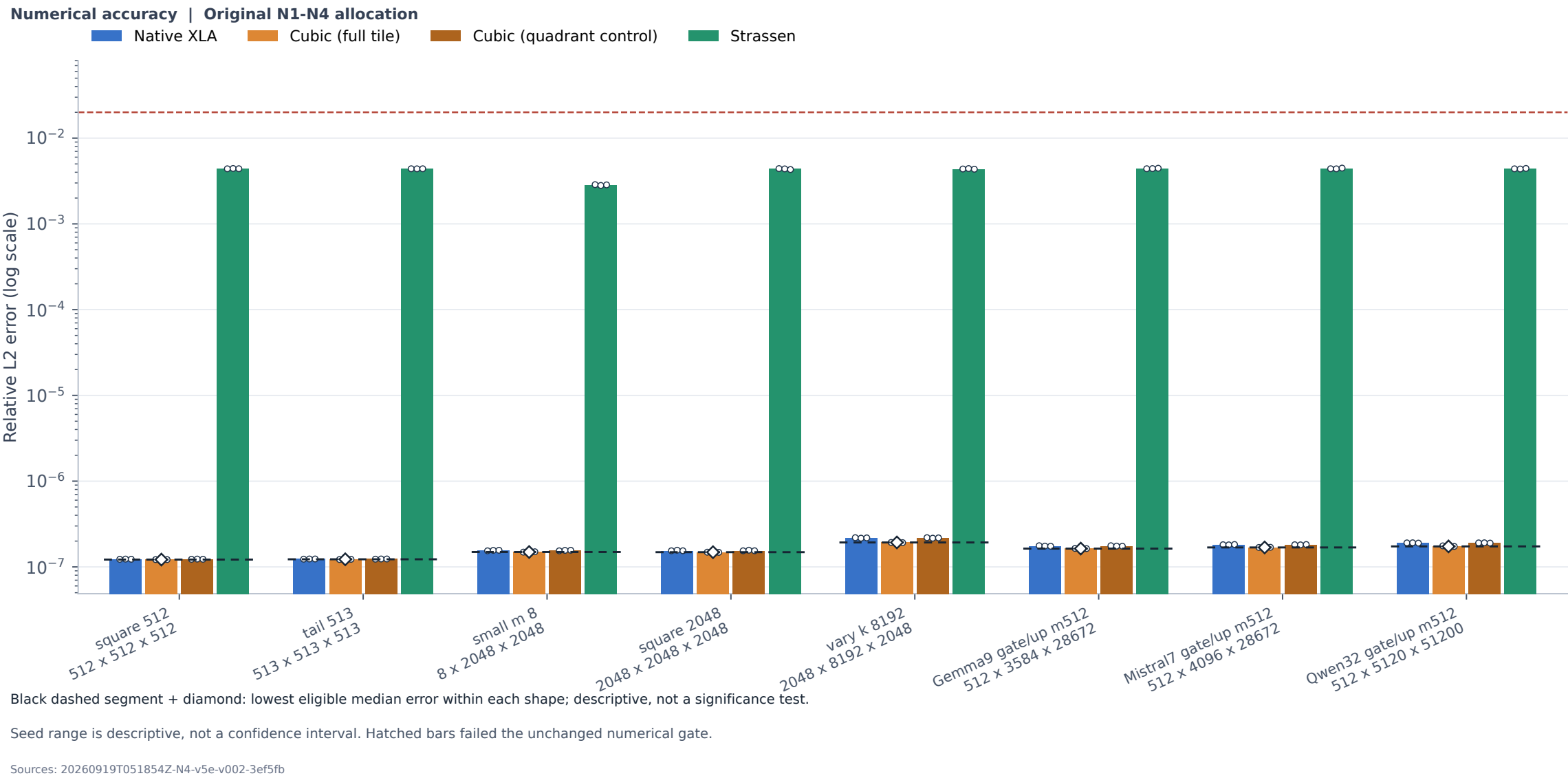
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Both changes 5, Product order 1. A lowest mean alone is not proof of a clear win.

OUR INTERPRETATION

These changes can improve the implementation, but the effect depends on shape and algorithm. Compare each variant with its own plain version. The fixed tile is 1024/1024/512; there is no Native XLA or full-tile Cubic baseline here. Product reordering alone does not prove that the TPU overlaps vector work and matrix multiplication.

N4 | gaussian - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



WHAT THIS GRAPH SHOWS

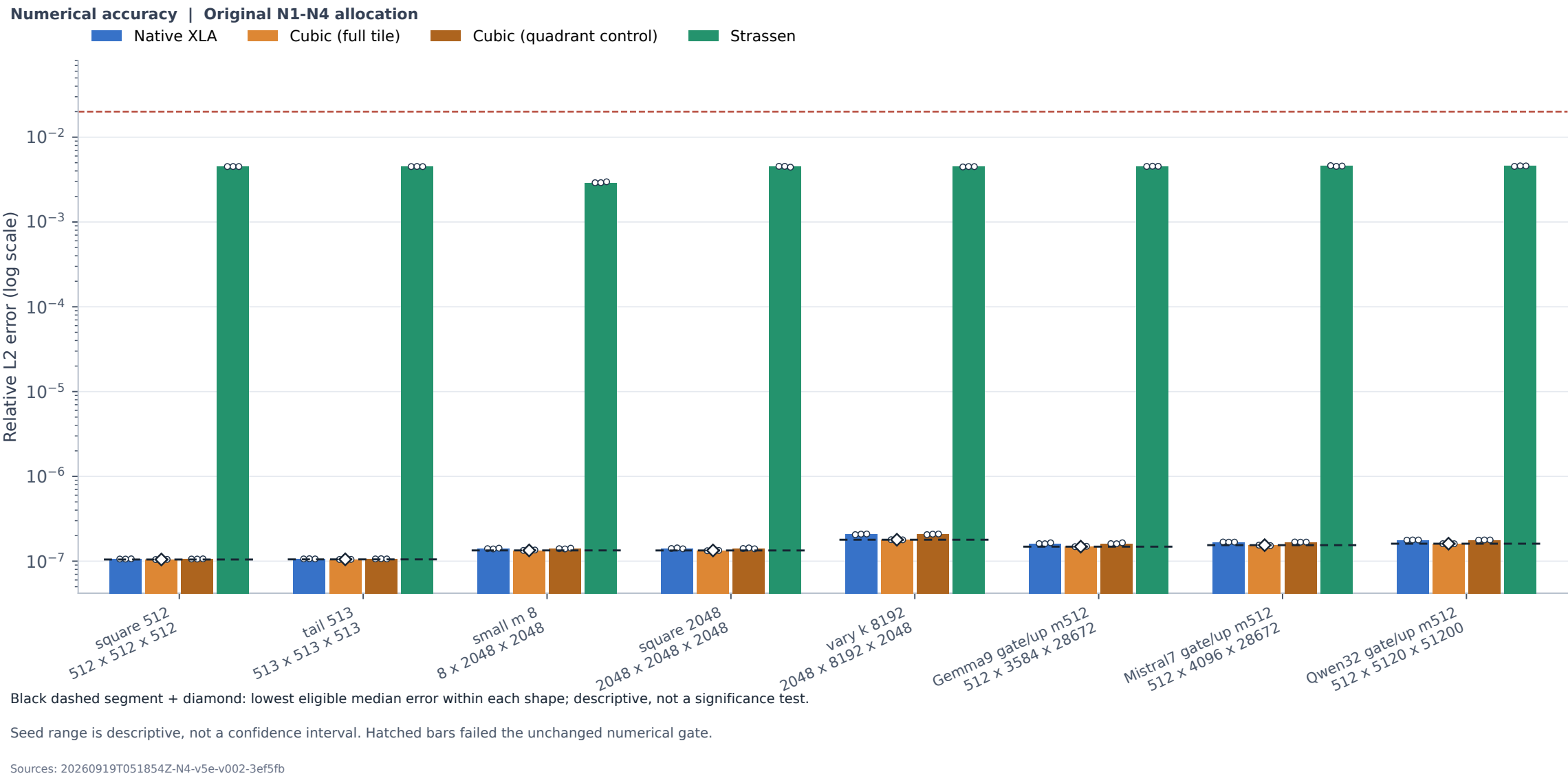
Bars show median relative L2 error across three seeds; dots and whiskers retain the individual seed values. Strassen's medians range from 0.00285 to 0.00441. 0 Strassen seed cases fail here. The red line is the 0.02 tolerance; the black segment marks the lowest eligible median error. No speed was measured.

OUR INTERPRETATION

Strassen introduces extra additions and subtractions and can have substantially larger numerical error. Passing a fixed tolerance is different from matching the accuracy of Cubic or Native. Performance gains must therefore be restricted to input conditions that have been checked; choosing a shape or tile alone cannot certify accuracy.

N4 | uniform - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



WHAT THIS GRAPH SHOWS

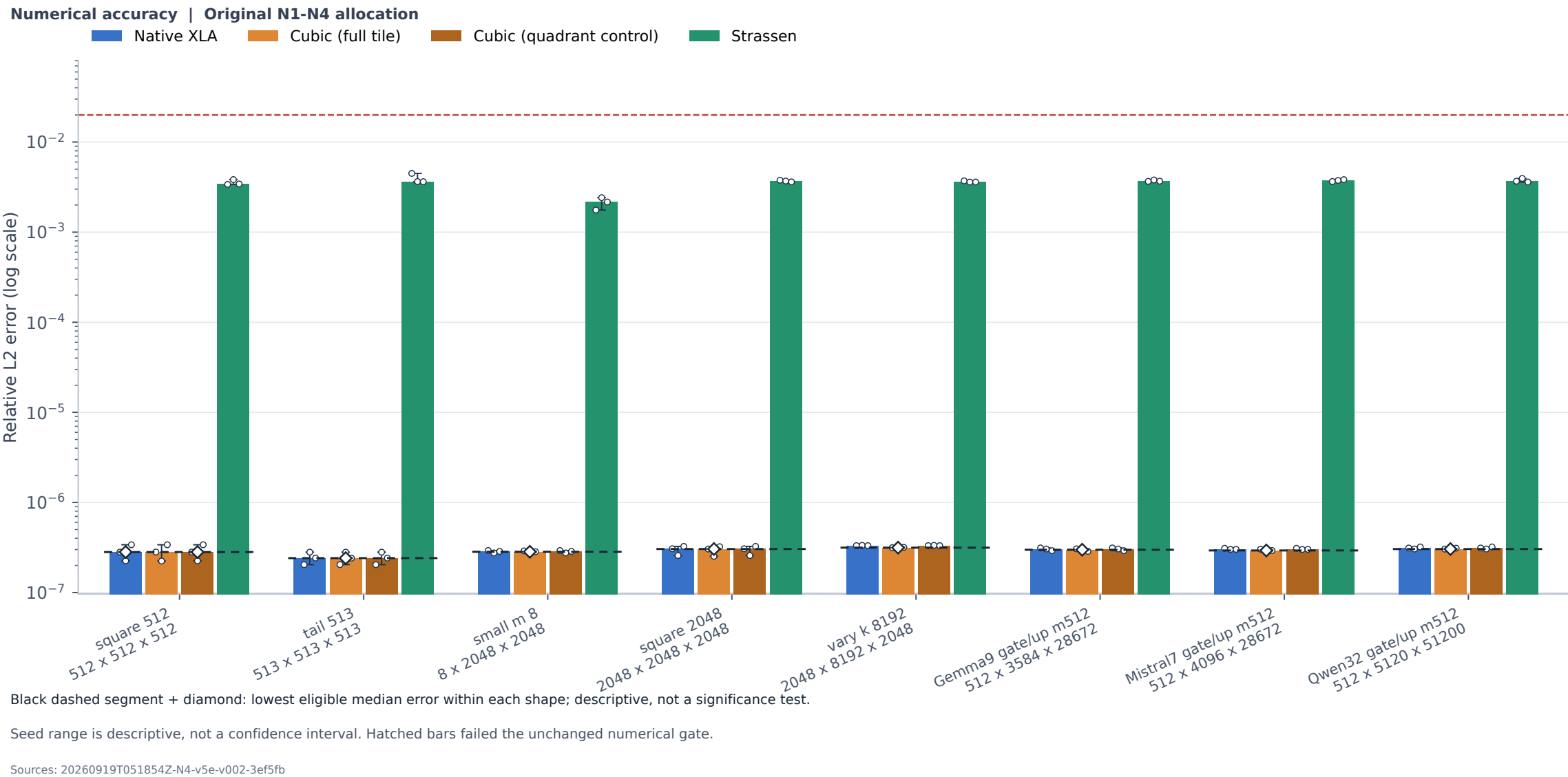
Bars show median relative L2 error across three seeds; dots and whiskers retain the individual seed values. Strassen's medians range from 0.00291 to 0.00459. 0 Strassen seed cases fail here. The red line is the 0.02 tolerance; the black segment marks the lowest eligible median error. No speed was measured.

OUR INTERPRETATION

Strassen introduces extra additions and subtractions and can have substantially larger numerical error. Passing a fixed tolerance is different from matching the accuracy of Cubic or Native. Performance gains must therefore be restricted to input conditions that have been checked; choosing a shape or tile alone cannot certify accuracy.

N4 | log scale - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



WHAT THIS GRAPH SHOWS

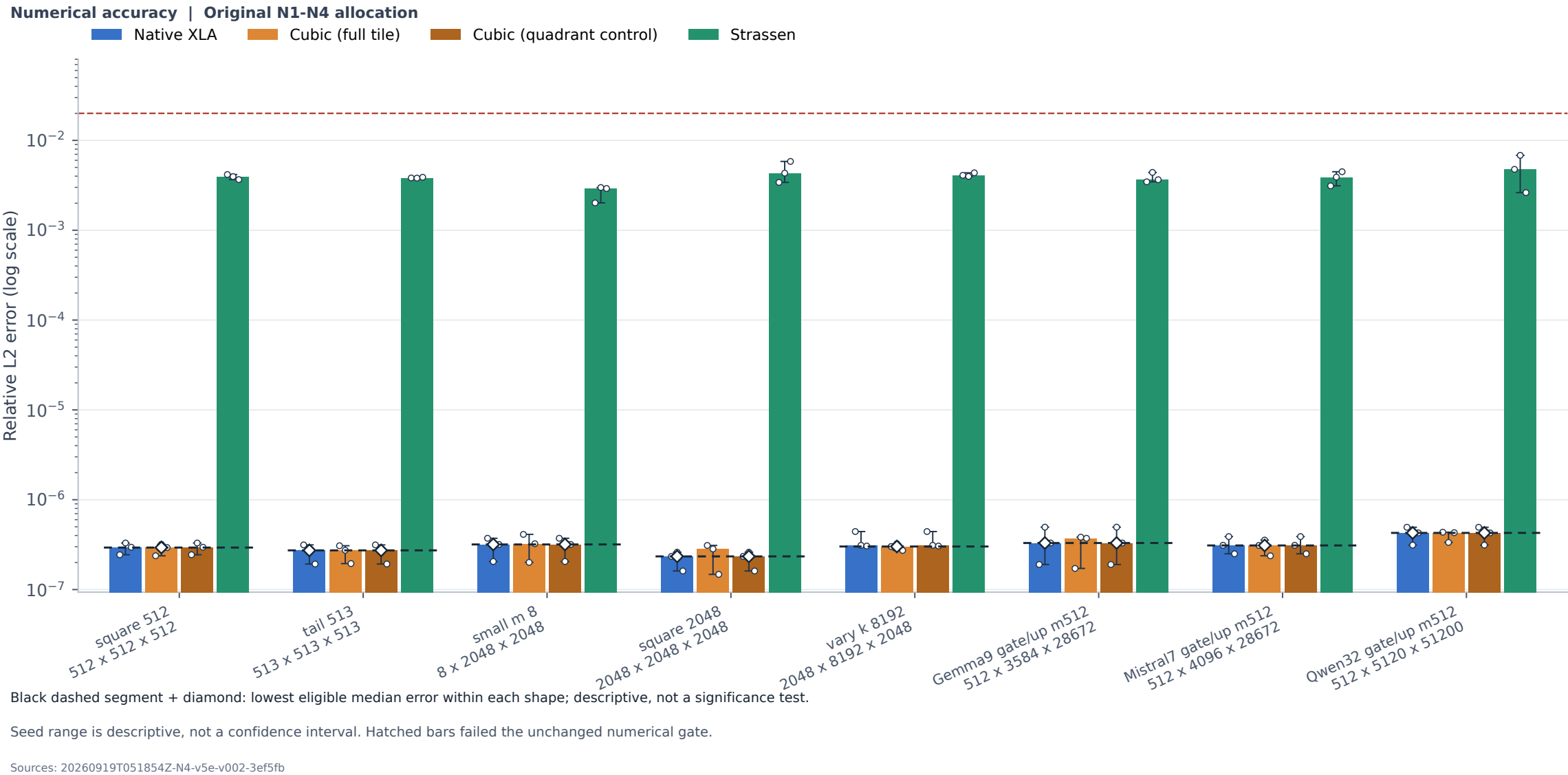
Bars show median relative L2 error across three seeds; dots and whiskers retain the individual seed values. Strassen's medians range from 0.00216 to 0.00375. 0 Strassen seed cases fail here. The red line is the 0.02 tolerance; the black segment marks the lowest eligible median error. No speed was measured.

OUR INTERPRETATION

Strassen introduces extra additions and subtractions and can have substantially larger numerical error. Passing a fixed tolerance is different from matching the accuracy of Cubic or Native. Performance gains must therefore be restricted to input conditions that have been checked; choosing a shape or tile alone cannot certify accuracy.

N4 | outliers - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



WHAT THIS GRAPH SHOWS

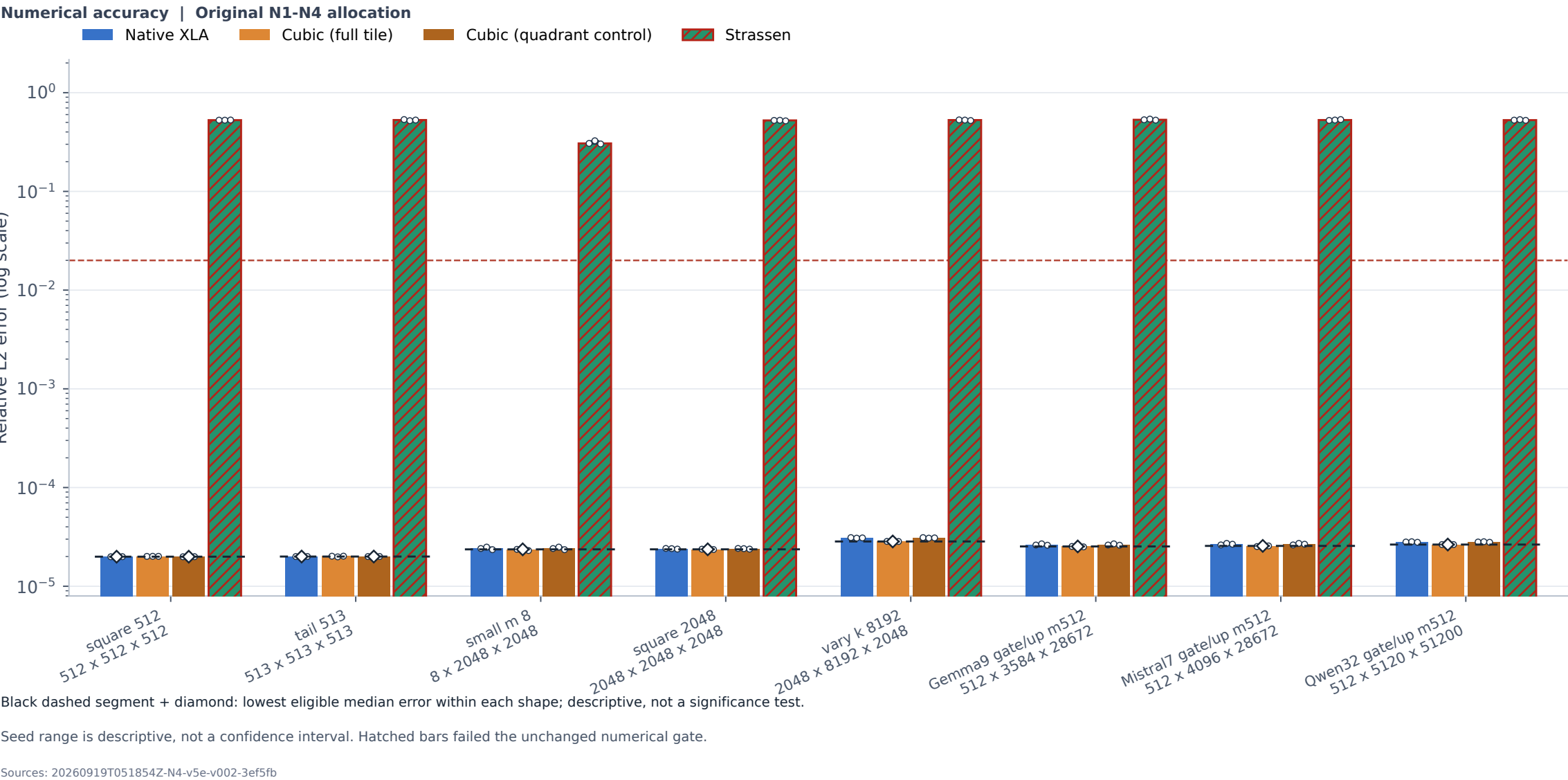
Bars show median relative L2 error across three seeds; dots and whiskers retain the individual seed values. Strassen's medians range from 0.00292 to 0.00476. 0 Strassen seed cases fail here. The red line is the 0.02 tolerance; the black segment marks the lowest eligible median error. No speed was measured.

OUR INTERPRETATION

Strassen introduces extra additions and subtractions and can have substantially larger numerical error. Passing a fixed tolerance is different from matching the accuracy of Cubic or Native. Performance gains must therefore be restricted to input conditions that have been checked; choosing a shape or tile alone cannot certify accuracy.

N4 | cancellation - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



WHAT THIS GRAPH SHOWS

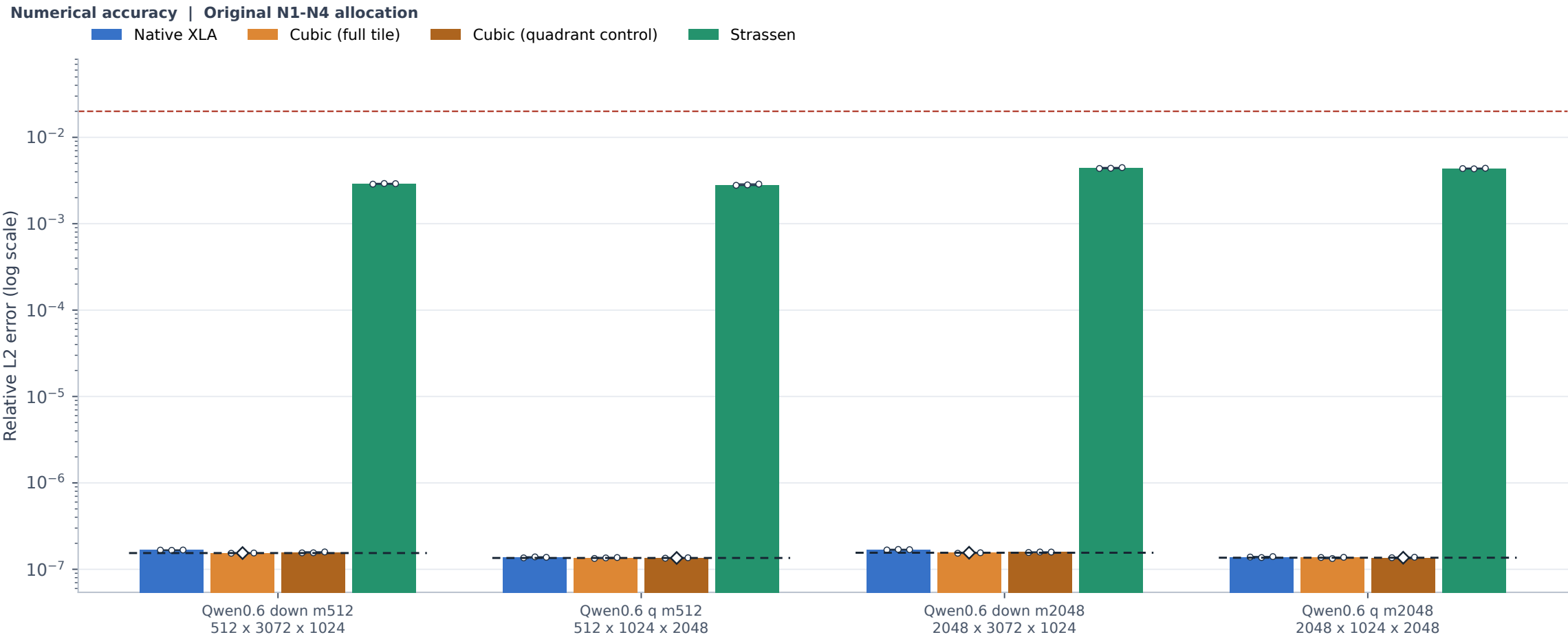
Bars show median relative L2 error across three seeds; dots and whiskers retain the individual seed values. Strassen's medians range from 0.306 to 0.531. 24 Strassen seed cases fail here. The red line is the 0.02 tolerance; the black segment marks the lowest eligible median error. No speed was measured.

OUR INTERPRETATION

All 24 Strassen cancellation cases fail, while the other algorithms pass. The median checked Strassen error is about 0.527, far above the 0.02 limit. This is a real input-sensitivity limit of the tested numerical contract; the speed plots cannot justify replacing Native or Cubic for arbitrary input values.

N4 | Actual Qwen weights - relative error

Accuracy only: median of 3 seeds; whiskers and dots show all seed values. No latency was measured.



Black dashed segment + diamond: lowest eligible median error within each shape; descriptive, not a significance test.

Seed range is descriptive, not a confidence interval. Hatched bars failed the unchanged numerical gate.

Sources: 20260919T052151Z-N4-real-weights-v5e-v003-cb86b9

WHAT THIS GRAPH SHOWS

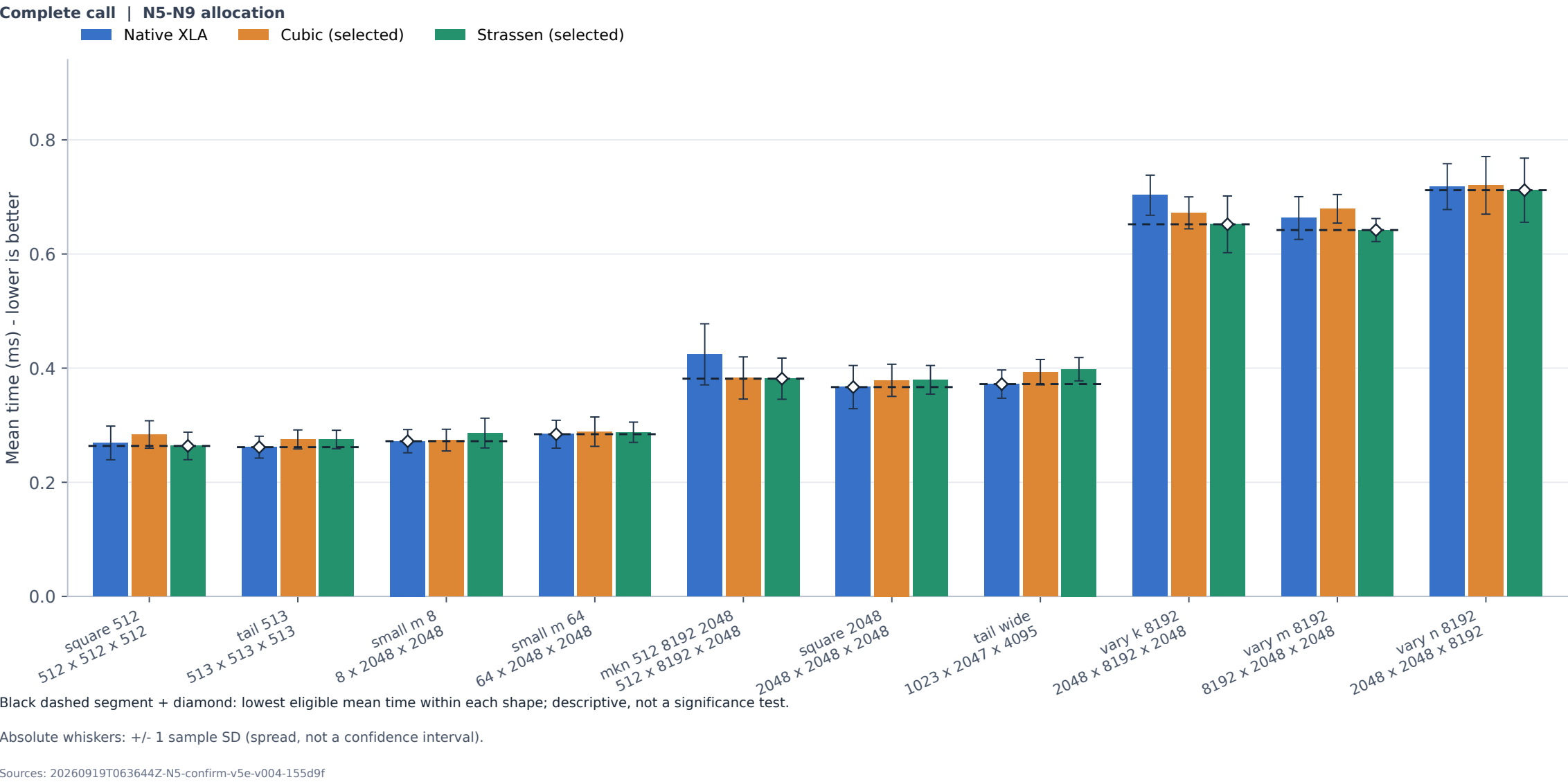
Bars show median relative L2 error across three seeds; dots and whiskers retain the individual seed values. Strassen's medians range from 0.00281 to 0.00439. 0 Strassen seed cases fail here. The red line is the 0.02 tolerance; the black segment marks the lowest eligible median error. No speed was measured.

OUR INTERPRETATION

All 48 cases using actual Qwen projection weights pass. However, the activations are synthetic Gaussian values. This checks compatibility with those weights, not real-model quality or arbitrary-input safety; the cancellation failures in the previous graph still apply.

N5 | Fresh confirmation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.



WHAT THIS GRAPH SHOWS

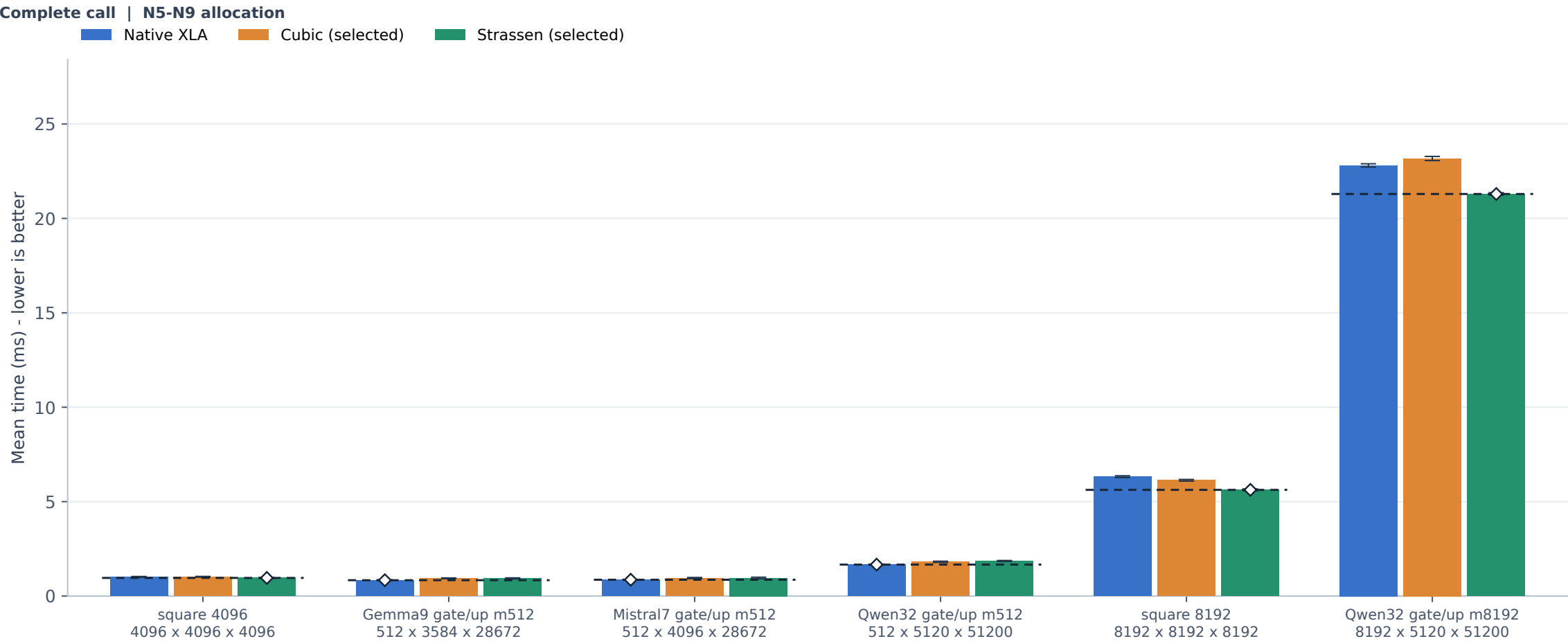
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 5, Native XLA 5. Archived paired intervals for Strassen versus Native: 3 faster, 3 slower, 4 inconclusive.

OUR INTERPRETATION

This is the fairer comparison: Cubic and Strassen each received 20 attempted tile/variant configurations per shape, then separate winners were frozen and measured on fresh data. Across 16 complete-call shapes, Strassen has 6 supported wins, 4 inconclusive comparisons and 6 losses versus Native; versus tuned Cubic it has 6 wins, 8 inconclusive and 2 losses. The benefit is selective, not universal.

N5 | Fresh confirmation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T063644Z-N5-confirm-v5e-v004-155d9f

WHAT THIS GRAPH SHOWS

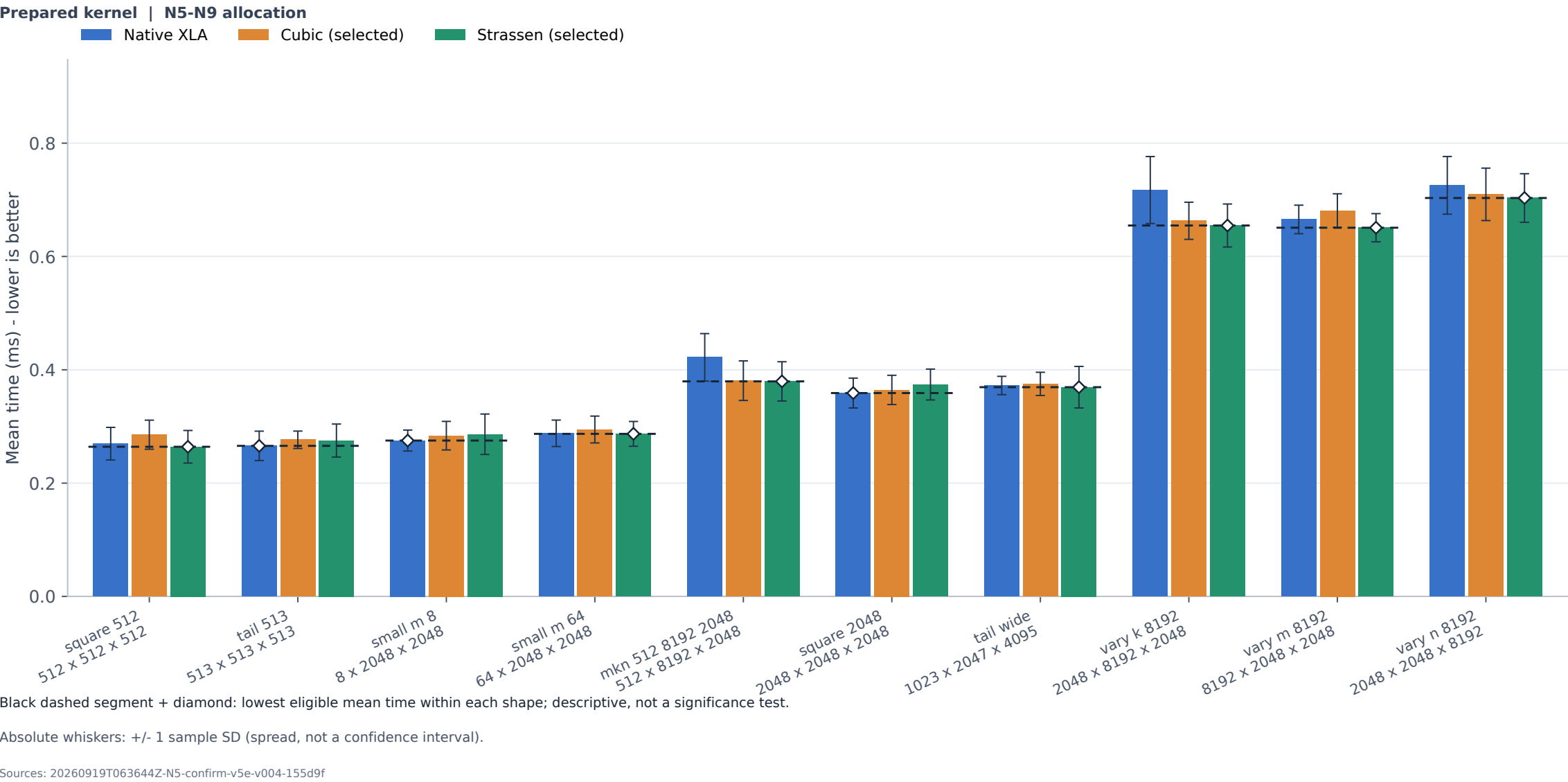
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 3, Native XLA 3. Archived paired intervals for Strassen versus Native: 3 faster, 3 slower, 0 inconclusive.

OUR INTERPRETATION

This is the fairer comparison: Cubic and Strassen each received 20 attempted tile/variant configurations per shape, then separate winners were frozen and measured on fresh data. Across 16 complete-call shapes, Strassen has 6 supported wins, 4 inconclusive comparisons and 6 losses versus Native; versus tuned Cubic it has 6 wins, 8 inconclusive and 2 losses. The benefit is selective, not universal.

N5 | Fresh confirmation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.



WHAT THIS GRAPH SHOWS

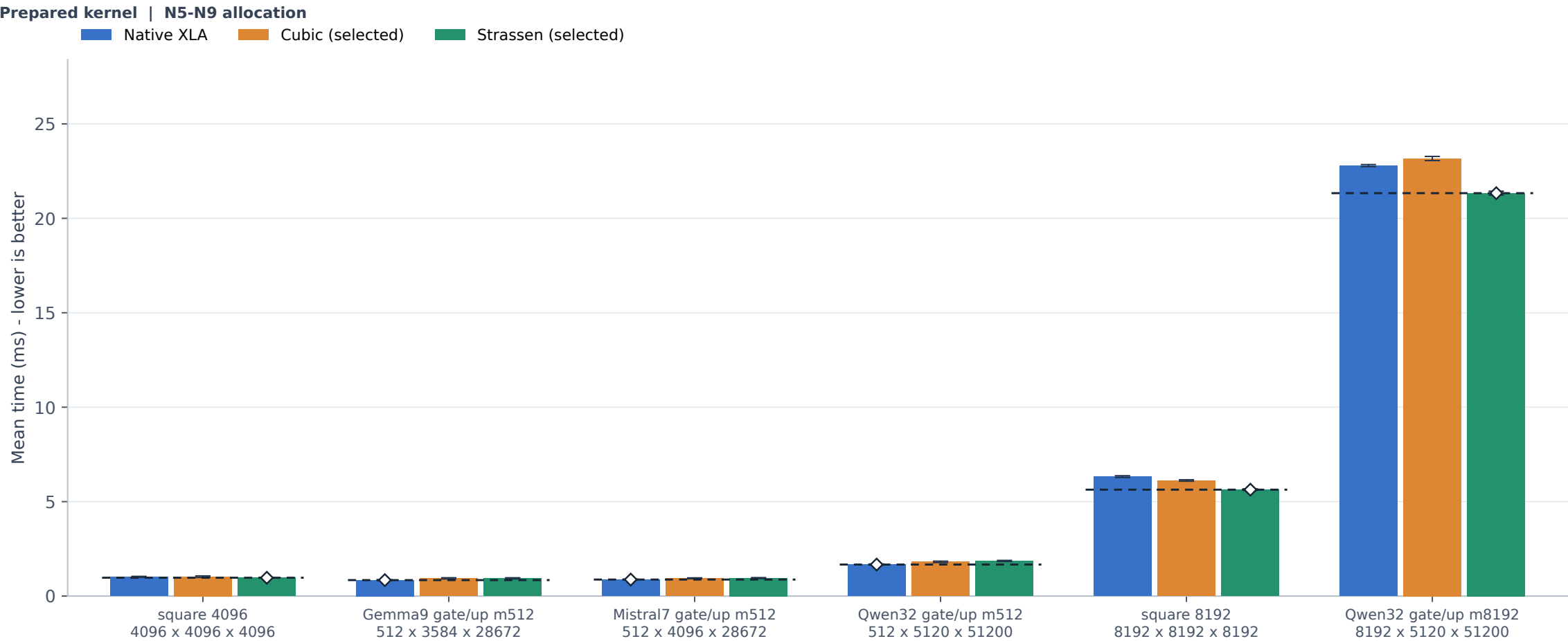
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 7, Native XLA 3. Archived paired intervals for Strassen versus Native: 4 faster, 1 slower, 5 inconclusive.

OUR INTERPRETATION

Prepared inputs change the practical comparison: across all 16 shapes, Strassen has 7 wins, 5 inconclusive comparisons and 4 losses versus Native, and 5 wins, 9 inconclusive and 2 losses versus tuned Cubic. The chosen configurations were selected by complete-call timing. Prepared performance assumes reusable prepared buffers and should not replace complete-call results.

N5 | Fresh confirmation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T063644Z-N5-confirm-v5e-v004-155d9f

WHAT THIS GRAPH SHOWS

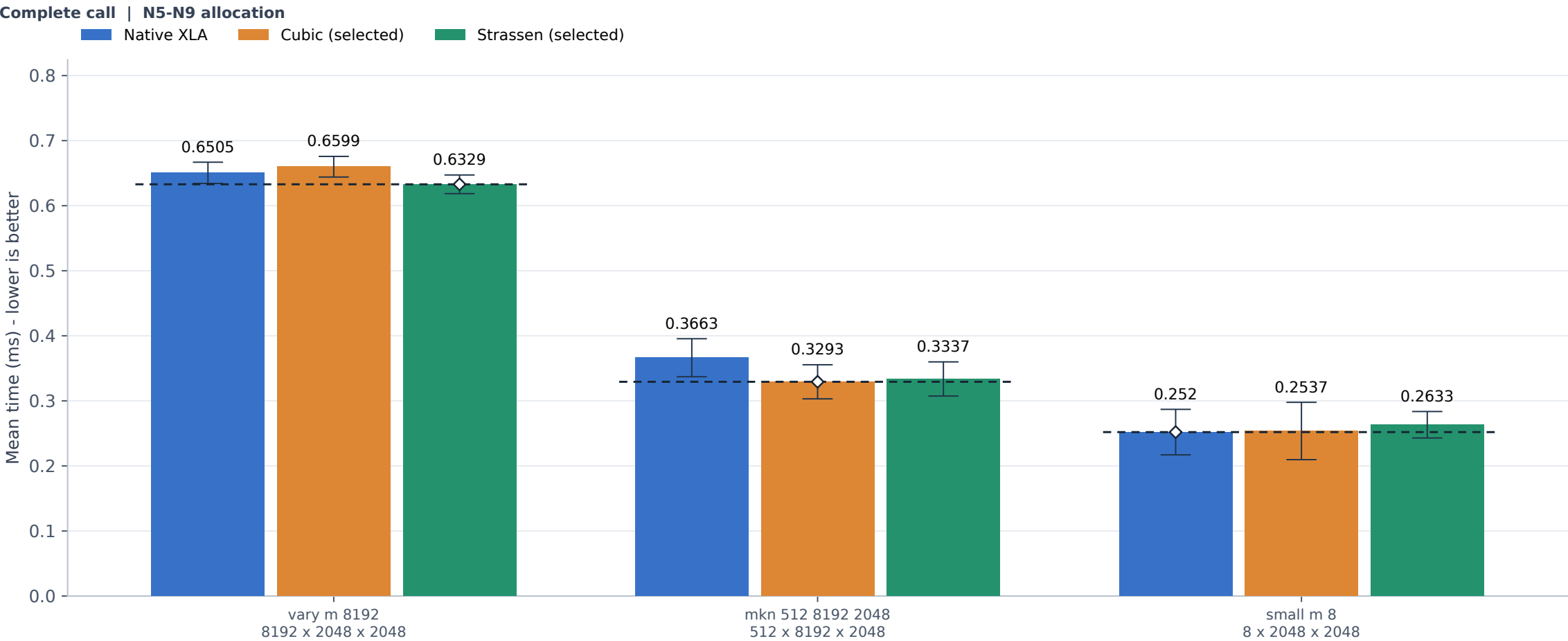
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 3, Native XLA 3. Archived paired intervals for Strassen versus Native: 3 faster, 3 slower, 0 inconclusive.

OUR INTERPRETATION

Prepared inputs change the practical comparison: across all 16 shapes, Strassen has 7 wins, 5 inconclusive comparisons and 4 losses versus Native, and 5 wins, 9 inconclusive and 2 losses versus tuned Cubic. The chosen configurations were selected by complete-call timing. Prepared performance assumes reusable prepared buffers and should not replace complete-call results.

N6 | Registered representatives - before profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T064743Z-N6-v5e-v004-e7bc5e

WHAT THIS GRAPH SHOWS

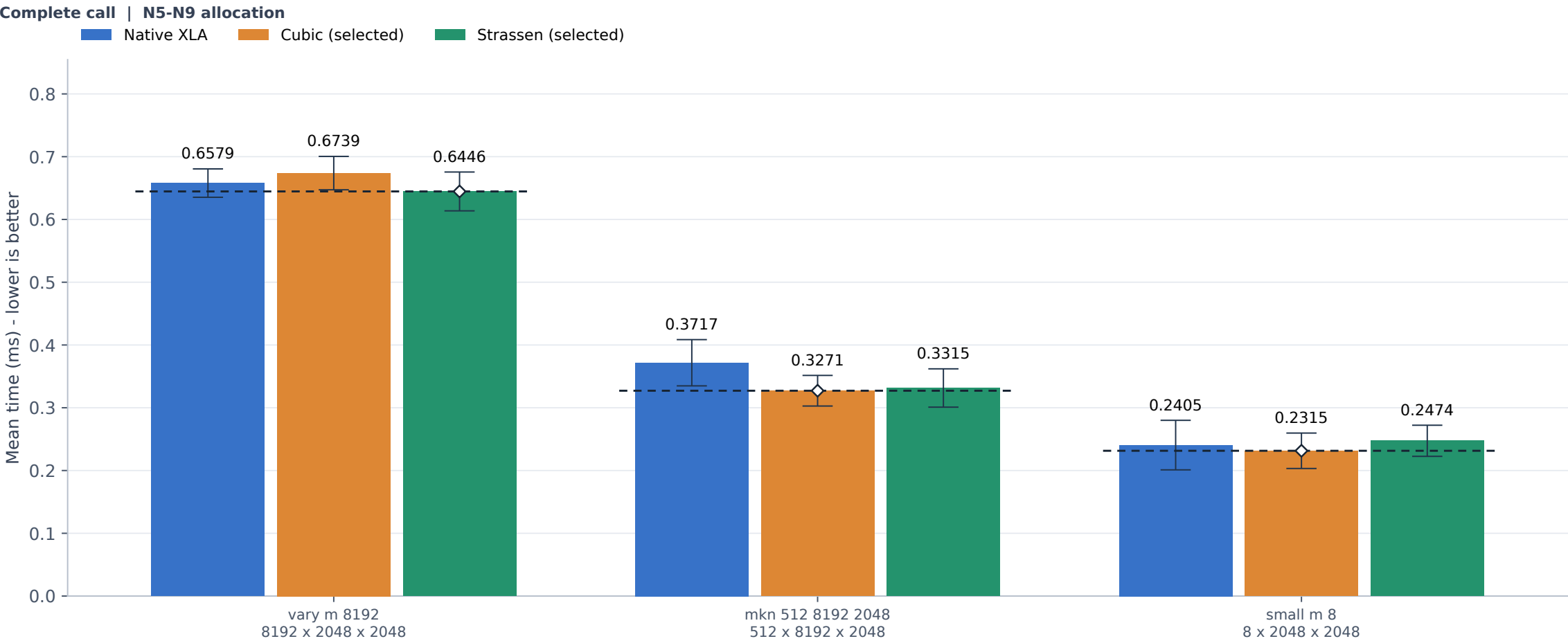
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 1, Cubic (selected) 1, Native XLA 1. Archived paired intervals for Strassen versus Native: 2 faster, 0 slower, 1 inconclusive.

OUR INTERPRETATION

Host-observed complete-call latency and profiled TPU-module duration answer different questions. Native has the shortest device module for all three original representatives, despite some different call-time rankings. These traces do not establish MXU utilization, vector overlap, or a measured decomposition of dispatch and memory costs.

N6 | Registered representatives - after profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T064743Z-N6-v5e-v004-e7bc5e

WHAT THIS GRAPH SHOWS

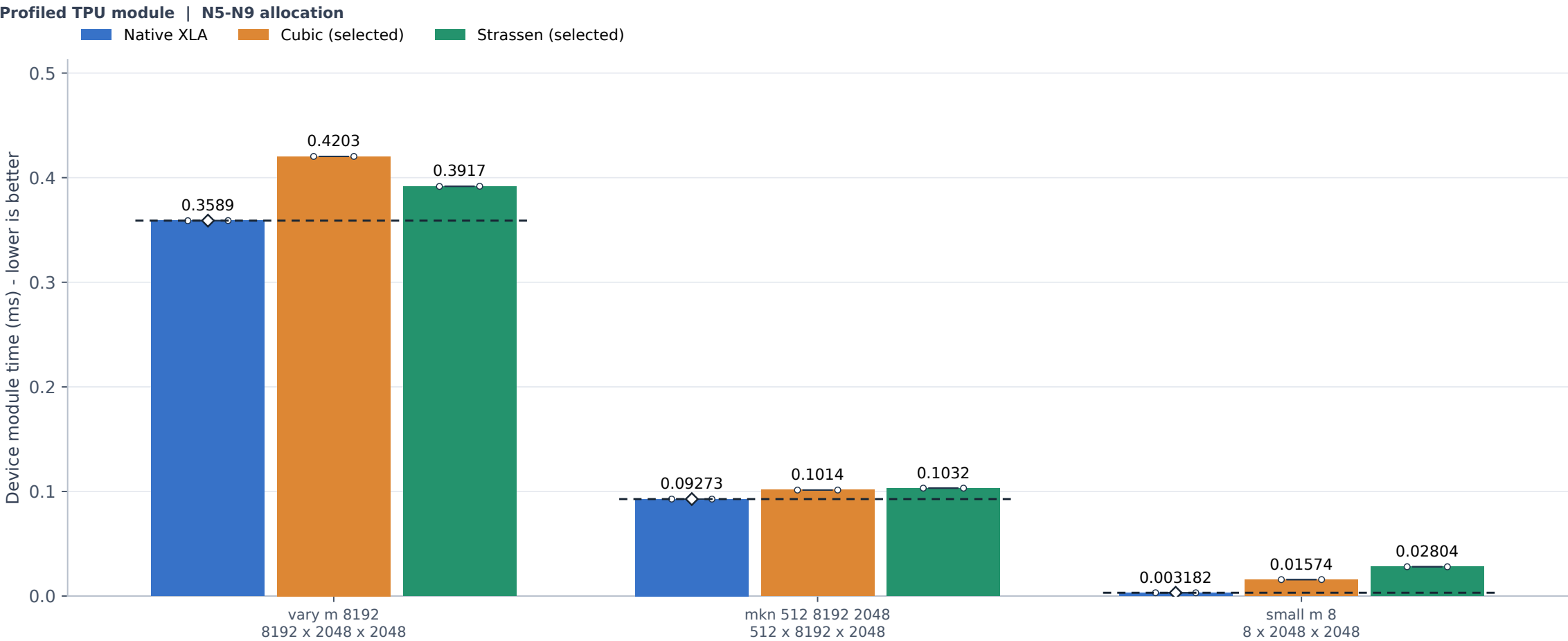
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 1, Cubic (selected) 2. Archived paired intervals for Strassen versus Native: 2 faster, 0 slower, 1 inconclusive.

OUR INTERPRETATION

Host-observed complete-call latency and profiled TPU-module duration answer different questions. Native has the shortest device module for all three original representatives, despite some different call-time rankings. These traces do not establish MXU utilization, vector overlap, or a measured decomposition of dispatch and memory costs.

N6 | Profiled TPU module

Mean of two 8-call captures; whiskers/dots show capture means. Profiler-instrumented, not ordinary call latency.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Capture-mean range is descriptive; no utilization, overlap or HBM-traffic claim.

Sources: 20260919T064743Z-N6-v5e-v004-e7bc5e

WHAT THIS GRAPH SHOWS

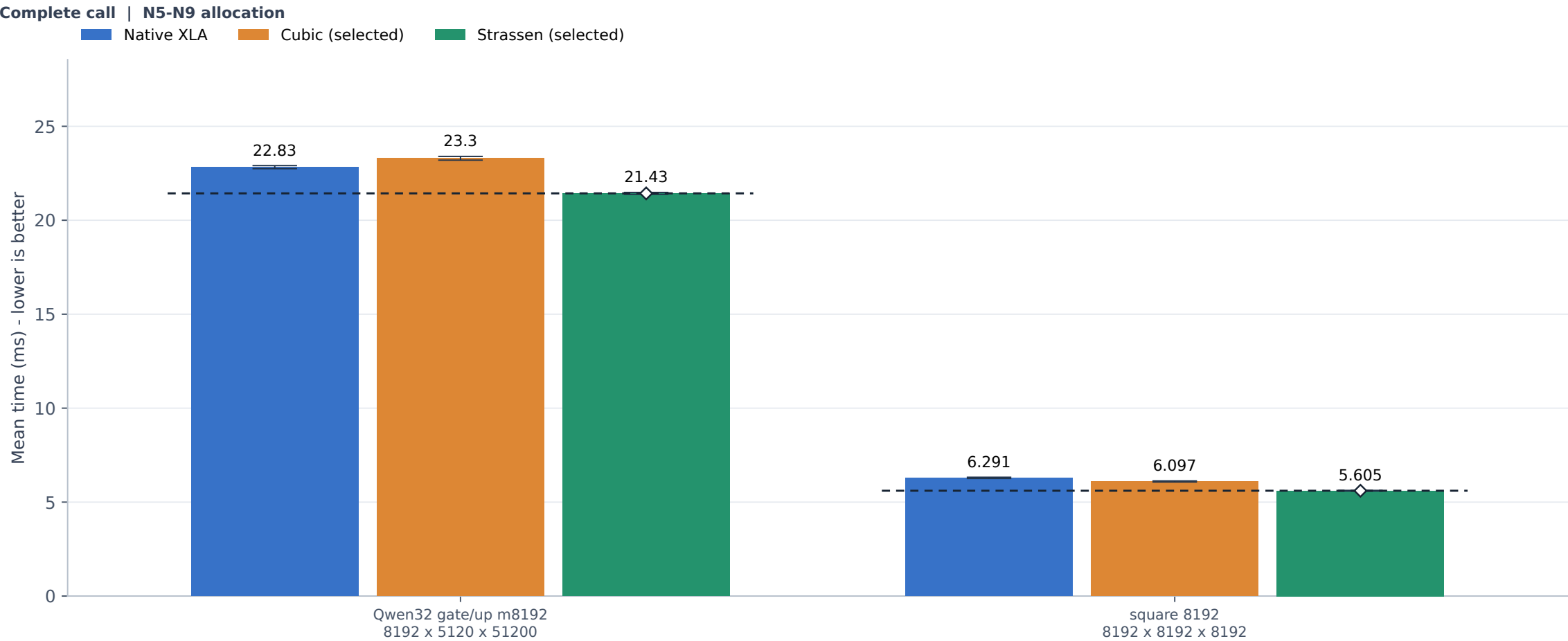
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 3. A lowest mean alone is not proof of a clear win.

OUR INTERPRETATION

Host-observed complete-call latency and profiled TPU-module duration answer different questions. Native has the shortest device module for all three original representatives, despite some different call-time rankings. These traces do not establish MXU utilization, vector overlap, or a measured decomposition of dispatch and memory costs.

N6 | Large-shape supplement - before profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T073356Z-N6-v5e-v004-dde5c0

WHAT THIS GRAPH SHOWS

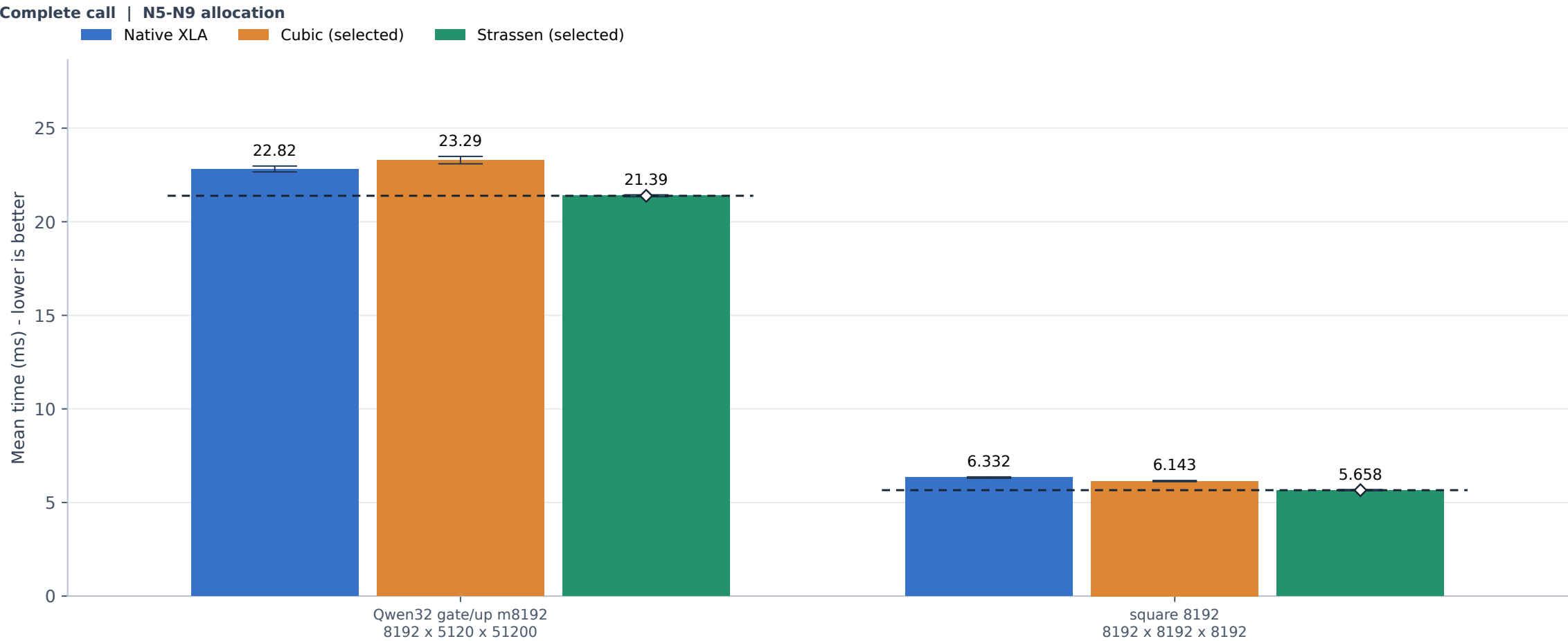
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2. Archived paired intervals for Strassen versus Native: 2 faster, 0 slower, 0 inconclusive.

OUR INTERPRETATION

The large Qwen and 8192-square cases also show a Strassen advantage inside the profiled device module: 1.061x and 1.129x versus Native. Before/after ordinary timings retain the same favorable direction. These shapes were chosen afterward from N5 winners, so this is a diagnostic of favorable cases, not an independent estimate of how often Strassen wins.

N6 | Large-shape supplement - after profiling

Ordinary synchronized host-observed latency; profiled device duration is a separate chart.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T073356Z-N6-v5e-v004-dde5c0

WHAT THIS GRAPH SHOWS

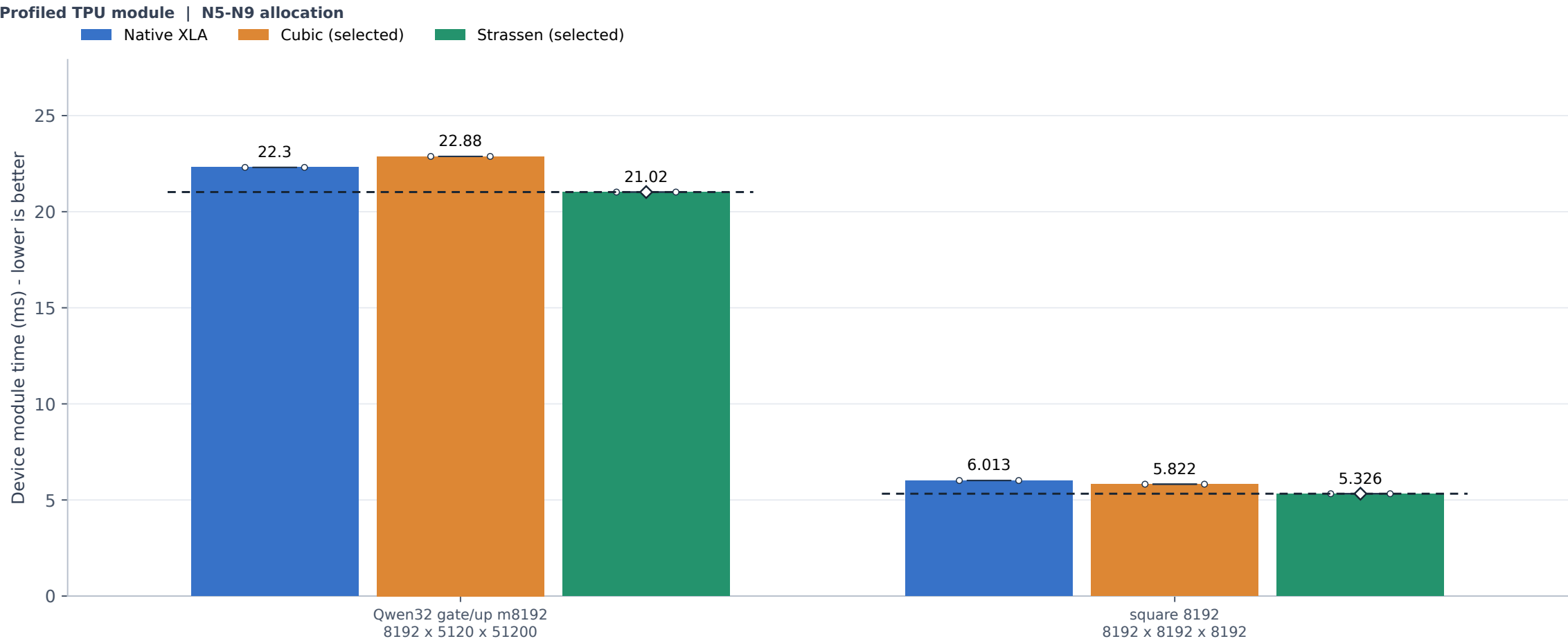
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2. Archived paired intervals for Strassen versus Native: 2 faster, 0 slower, 0 inconclusive.

OUR INTERPRETATION

The large Qwen and 8192-square cases also show a Strassen advantage inside the profiled device module: 1.061x and 1.129x versus Native. Before/after ordinary timings retain the same favorable direction. These shapes were chosen afterward from N5 winners, so this is a diagnostic of favorable cases, not an independent estimate of how often Strassen wins.

N6 | Profiled TPU module - large-shape supplement

Mean of two 8-call captures; whiskers/dots show capture means. Profiler-instrumented, not ordinary call latency.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Capture-mean range is descriptive; no utilization, overlap or HBM-traffic claim.

Sources: 20260919T073356Z-N6-v5e-v004-dde5c0

WHAT THIS GRAPH SHOWS

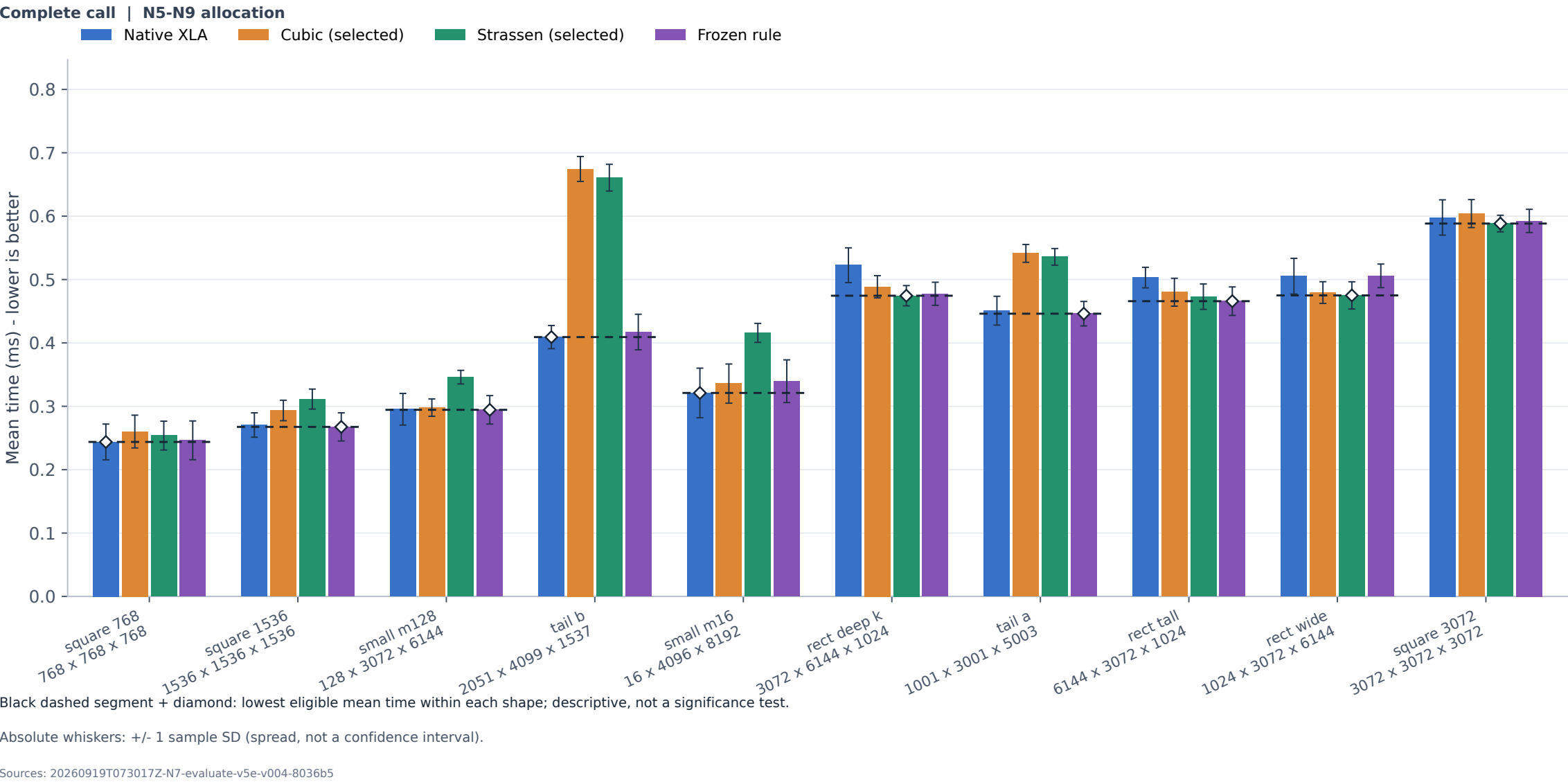
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2. A lowest mean alone is not proof of a clear win.

OUR INTERPRETATION

The large Qwen and 8192-square cases also show a Strassen advantage inside the profiled device module: 1.061x and 1.129x versus Native. Before/after ordinary timings retain the same favorable direction. These shapes were chosen afterward from N5 winners, so this is a diagnostic of favorable cases, not an independent estimate of how often Strassen wins.

N7 | Frozen-rule evaluation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.



WHAT THIS GRAPH SHOWS

The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 3, Frozen rule 4, Strassen (selected) 3. Archived paired intervals for the independently tuned Strassen alternative versus Native: 3 faster, 5 slower, 2 inconclusive.

OUR INTERPRETATION

The rule was frozen from N5 before these held-out shapes were measured. It chooses Native on 13 shapes, Strassen on 2 and Cubic on 1. All 3 custom routes beat Native in complete-call timing, and those gains repeat on a fresh v5e allocation. Independent comparator tuning did not update the rule. Duplicate Native/rule bars can differ through timing noise rather than algorithmic improvement.

N7 | Frozen-rule evaluation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

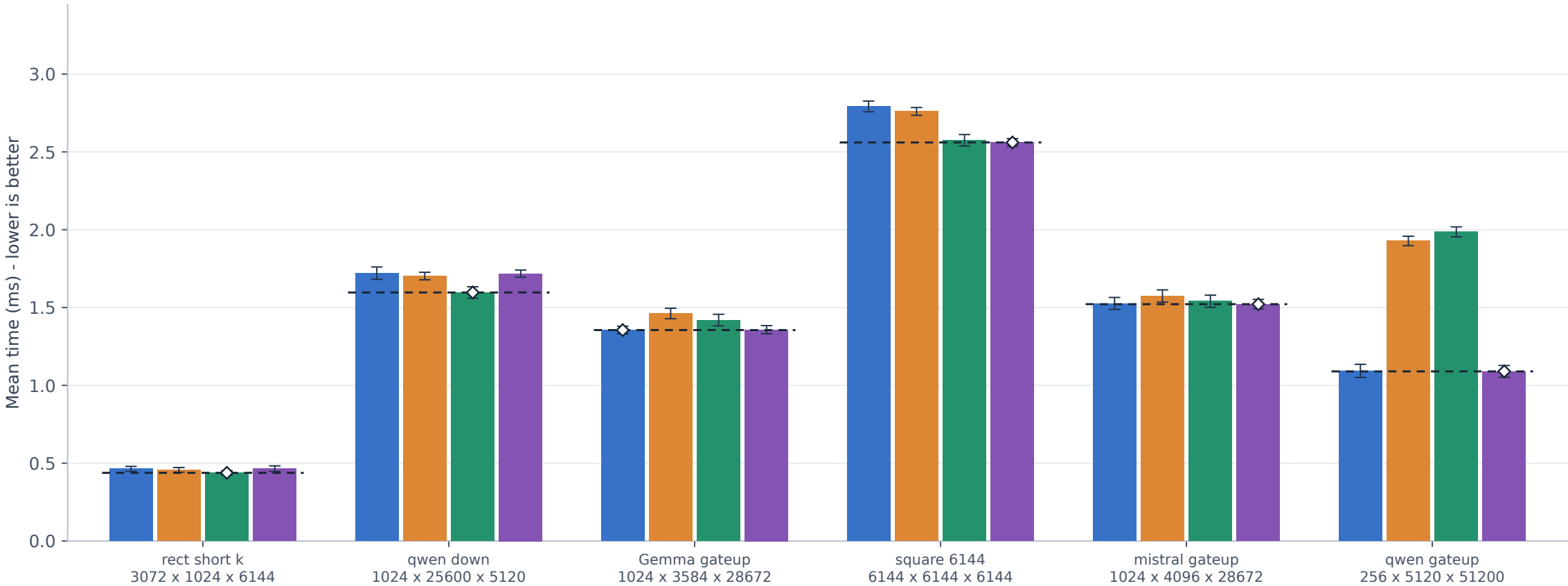
Complete call | N5-N9 allocation

Native XLA

Cubic (selected)

Strassen (selected)

Frozen rule



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T073017Z-N7-evaluate-v5e-v004-8036b5

WHAT THIS GRAPH SHOWS

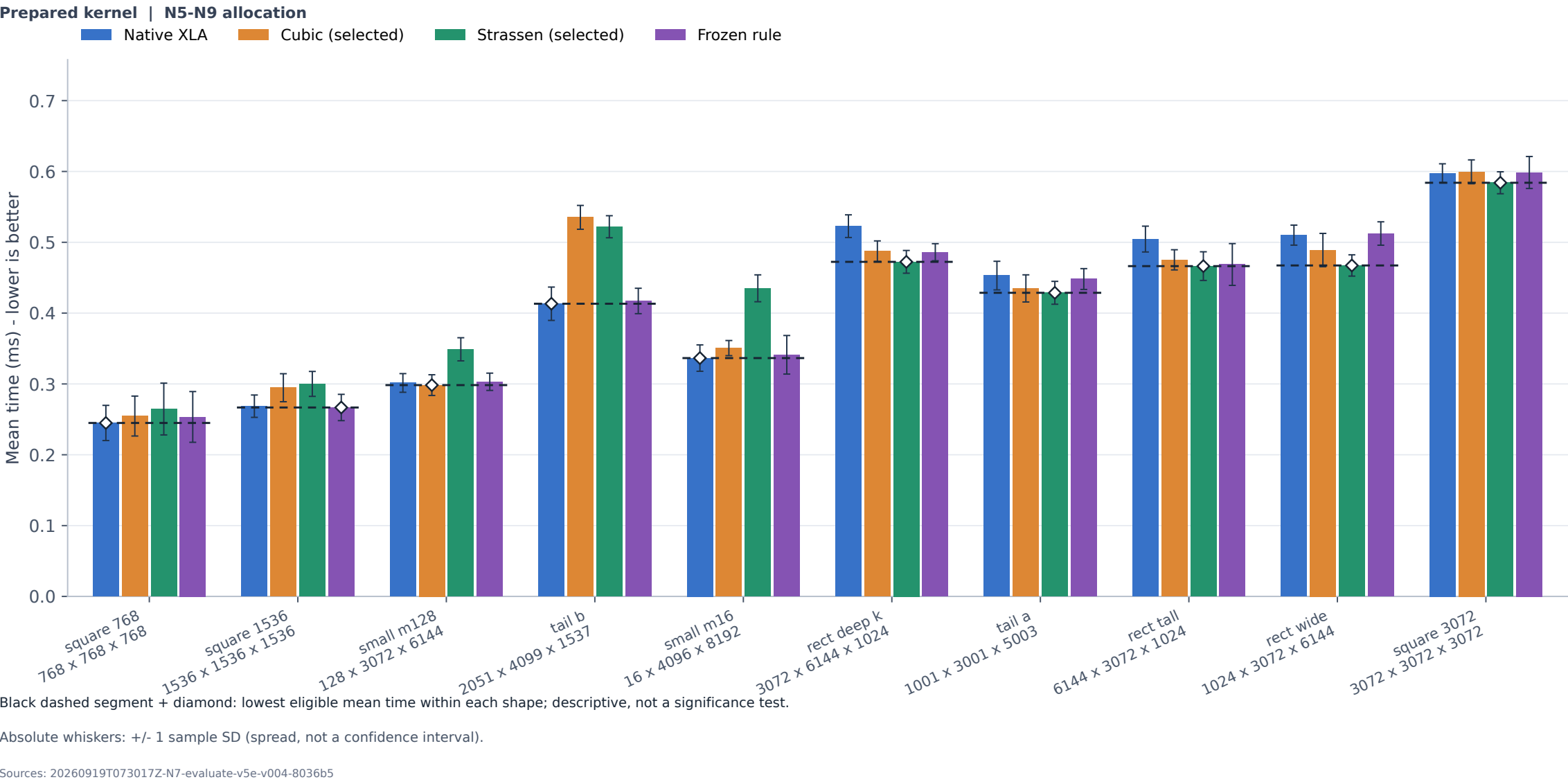
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2, Native XLA 1, Frozen rule 3. Archived paired intervals for the independently tuned Strassen alternative versus Native: 3 faster, 2 slower, 1 inconclusive.

OUR INTERPRETATION

The rule was frozen from N5 before these held-out shapes were measured. It chooses Native on 13 shapes, Strassen on 2 and Cubic on 1. All 3 custom routes beat Native in complete-call timing, and those gains repeat on a fresh v5e allocation. Independent comparator tuning did not update the rule. Duplicate Native/rule bars can differ through timing noise rather than algorithmic improvement.

N7 | Frozen-rule evaluation - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.



WHAT THIS GRAPH SHOWS

The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 3, Frozen rule 1, Cubic (selected) 1, Strassen (selected) 5. Archived paired intervals for the independently tuned Strassen alternative versus Native: 5 faster, 5 slower, 0 inconclusive.

OUR INTERPRETATION

The rule was frozen from N5 before these held-out shapes were measured. It chooses Native on 13 shapes, Strassen on 2 and Cubic on 1. All 3 custom routes beat Native in complete-call timing, and those gains repeat on a fresh v5e allocation. Independent comparator tuning did not update the rule. Duplicate Native/rule bars can differ through timing noise rather than algorithmic improvement.

N7 | Frozen-rule evaluation - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

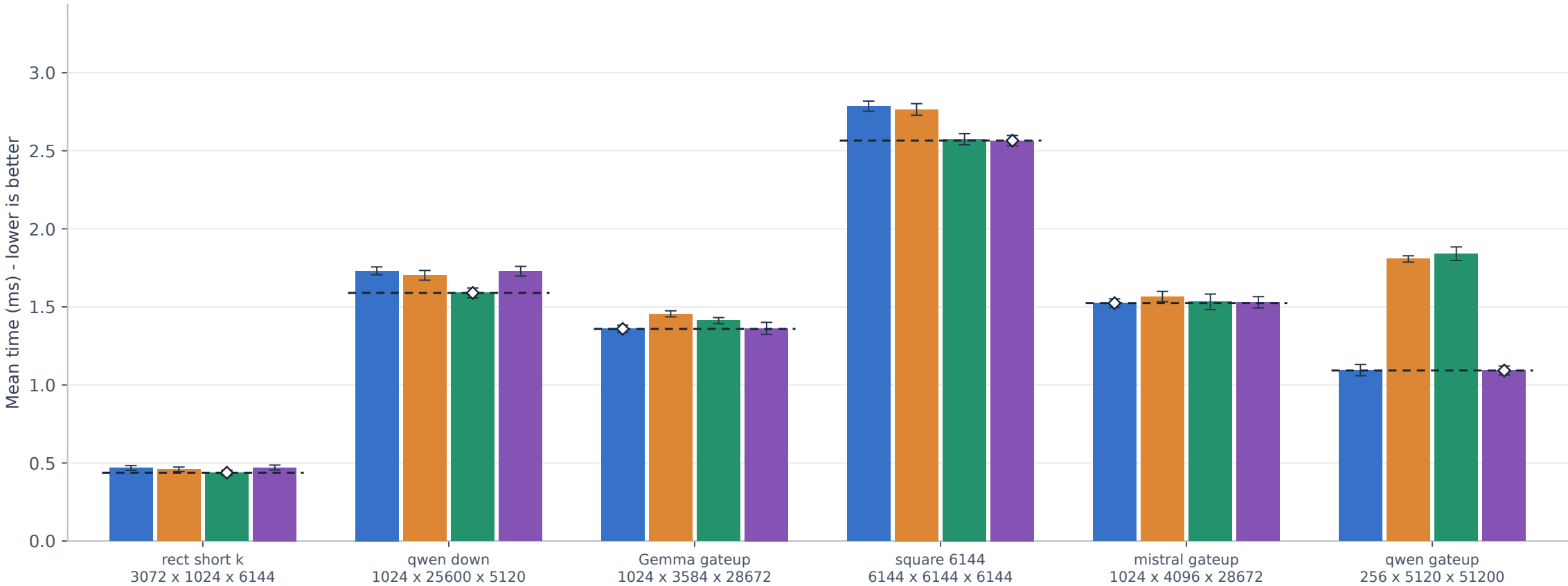
Prepared kernel | N5-N9 allocation

Native XLA

Cubic (selected)

Strassen (selected)

Frozen rule



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T073017Z-N7-evaluate-v5e-v004-8036b5

WHAT THIS GRAPH SHOWS

The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2, Native XLA 2, Frozen rule 2. Archived paired intervals for the independently tuned Strassen alternative versus Native: 3 faster, 2 slower, 1 inconclusive.

OUR INTERPRETATION

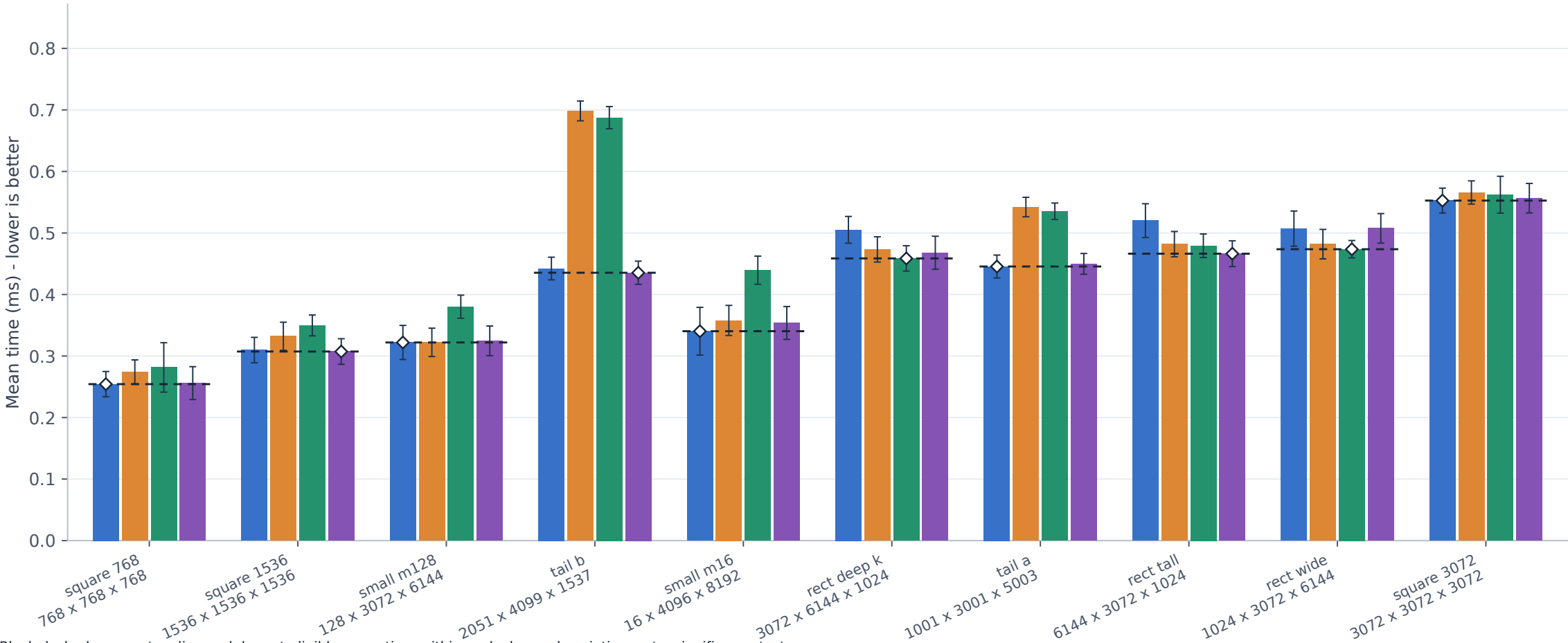
The rule was frozen from N5 before these held-out shapes were measured. It chooses Native on 13 shapes, Strassen on 2 and Cubic on 1. All 3 custom routes beat Native in complete-call timing, and those gains repeat on a fresh v5e allocation. Independent comparator tuning did not update the rule. Duplicate Native/rule bars can differ through timing noise rather than algorithmic improvement.

N7 | Fresh-v5e replication - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | Fresh v5e replication

Native XLA Cubic (selected) Strassen (selected) Frozen rule



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T084449Z-N7-replicate-v5e-v004-955768

WHAT THIS GRAPH SHOWS

The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 5, Frozen rule 3, Strassen (selected) 2. Archived paired intervals for the independently tuned Strassen alternative versus Native: 3 faster, 7 slower, 0 inconclusive.

OUR INTERPRETATION

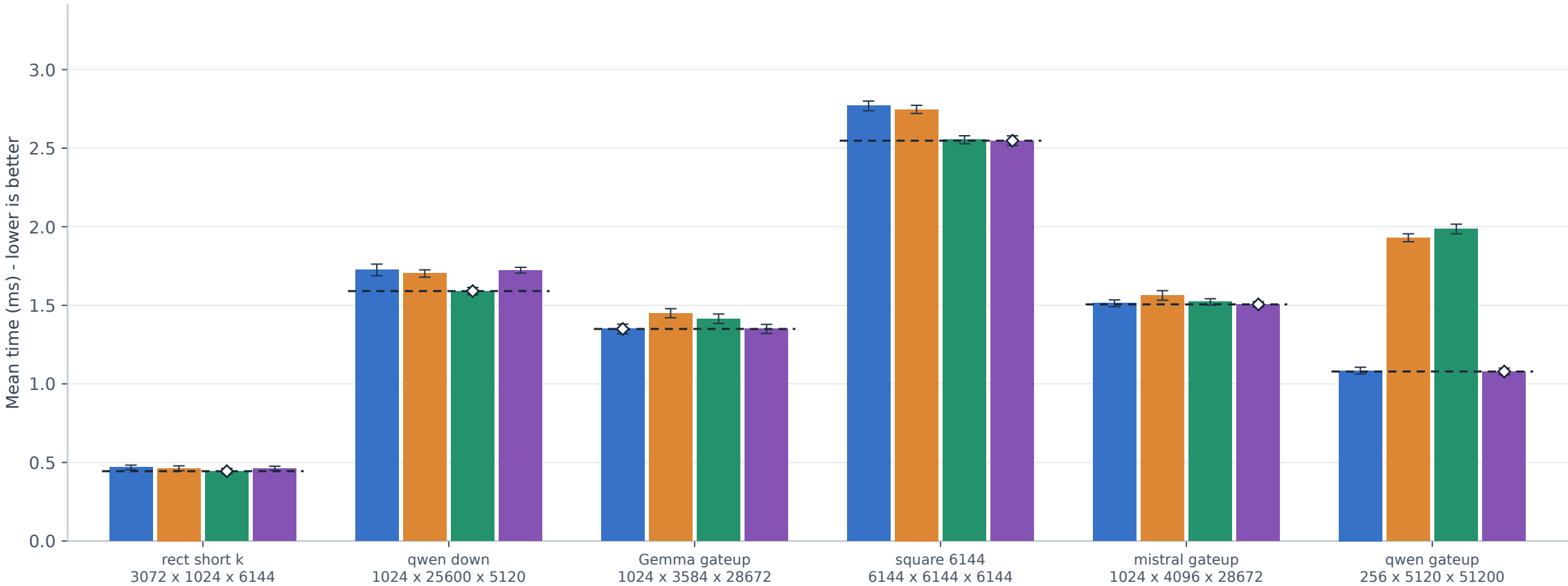
No retuning or rule update occurred on this fresh v5e allocation. The same three custom complete-call gains repeat: deep-K Cubic 1.080x, tall Strassen 1.115x and square Strassen 1.087x versus Native. A nominal extra Native-versus-Native win is timing variability, not a new algorithmic gain. This is fresh-v5e replication; v6e remains future work.

N7 | Fresh-v5e replication - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Complete call | Fresh v5e replication

Native XLA Cubic (selected) Strassen (selected) Frozen rule



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T084449Z-N7-replicate-v5e-v004-955768

WHAT THIS GRAPH SHOWS

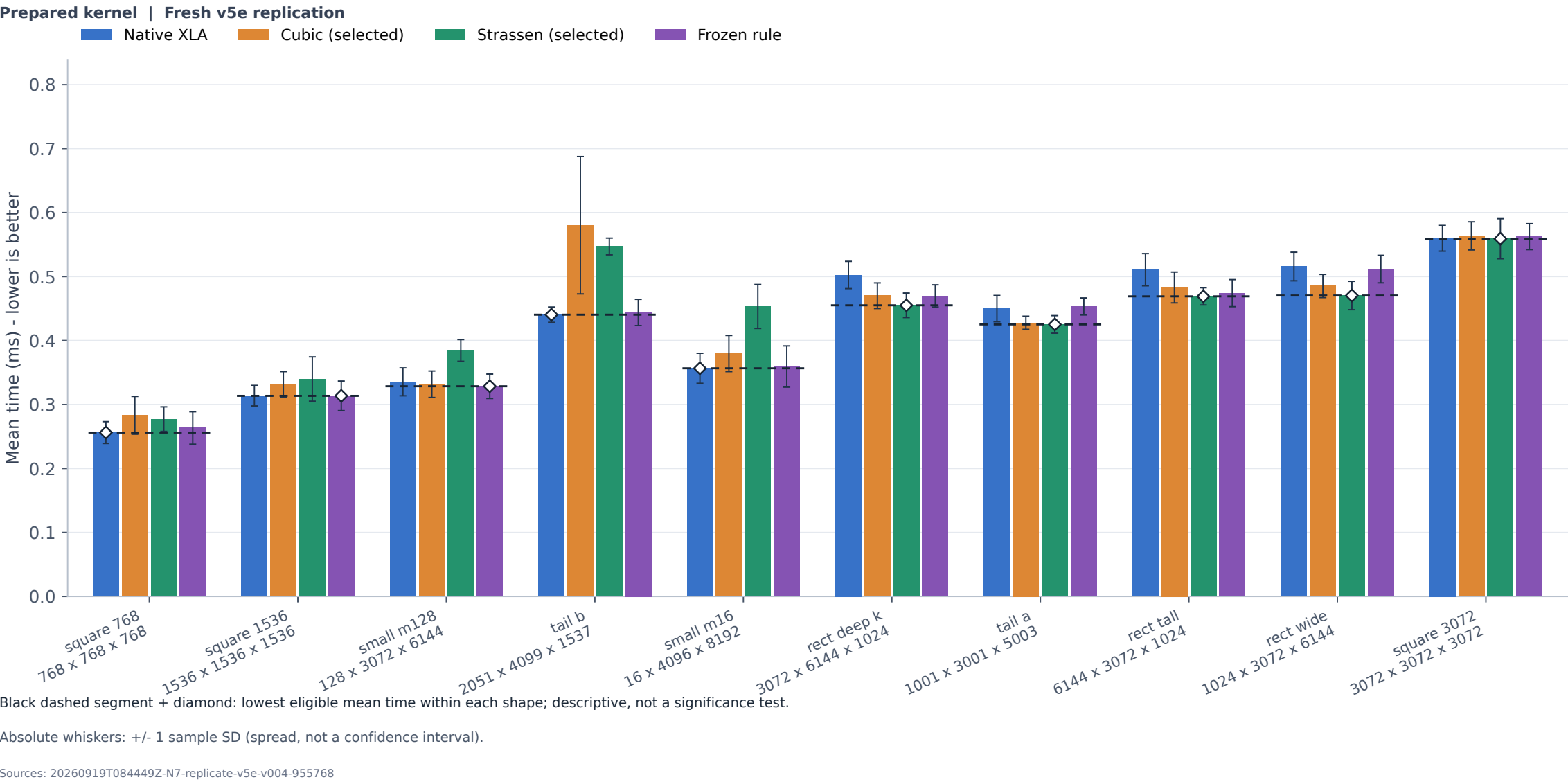
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2, Native XLA 1, Frozen rule 3. Archived paired intervals for the independently tuned Strassen alternative versus Native: 3 faster, 2 slower, 1 inconclusive.

OUR INTERPRETATION

No retuning or rule update occurred on this fresh v5e allocation. The same three custom complete-call gains repeat: deep-K Cubic 1.080x, tall Strassen 1.115x and square Strassen 1.087x versus Native. A nominal extra Native-versus-Native win is timing variability, not a new algorithmic gain. This is fresh-v5e replication; v6e remains future work.

N7 | Fresh-v5e replication - shapes 1-10

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.



WHAT THIS GRAPH SHOWS

The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native XLA 3, Frozen rule 2, Strassen (selected) 5. Archived paired intervals for the independently tuned Strassen alternative versus Native: 4 faster, 5 slower, 1 inconclusive.

OUR INTERPRETATION

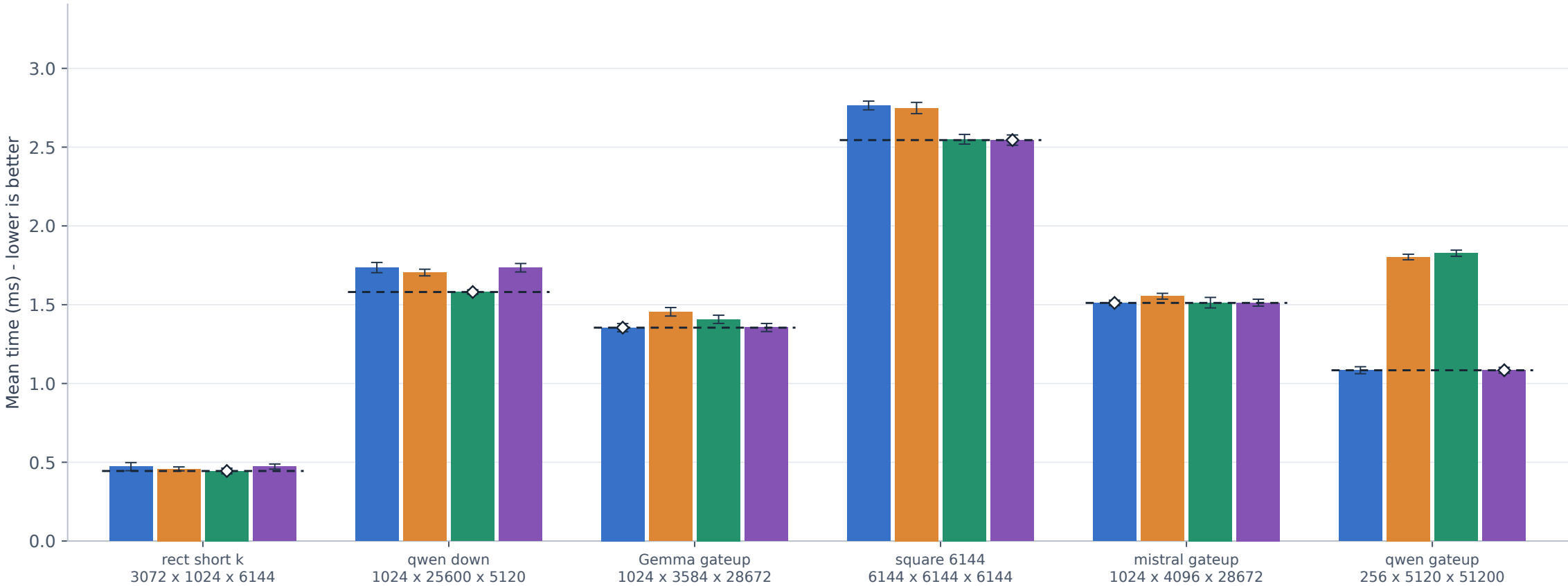
No retuning or rule update occurred on this fresh v5e allocation. The same three custom complete-call gains repeat: deep-K Cubic 1.080x, tall Strassen 1.115x and square Strassen 1.087x versus Native. A nominal extra Native-versus-Native win is timing variability, not a new algorithmic gain. This is fresh-v5e replication; v6e remains future work.

N7 | Fresh-v5e replication - shapes 11-16

30 paired rounds; independent family selections. Matrix dimensions are M x K x N.

Prepared kernel | Fresh v5e replication

Native XLA Cubic (selected) Strassen (selected) Frozen rule



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Absolute whiskers: +/- 1 sample SD (spread, not a confidence interval).

Sources: 20260919T084449Z-N7-replicate-v5e-v004-955768

WHAT THIS GRAPH SHOWS

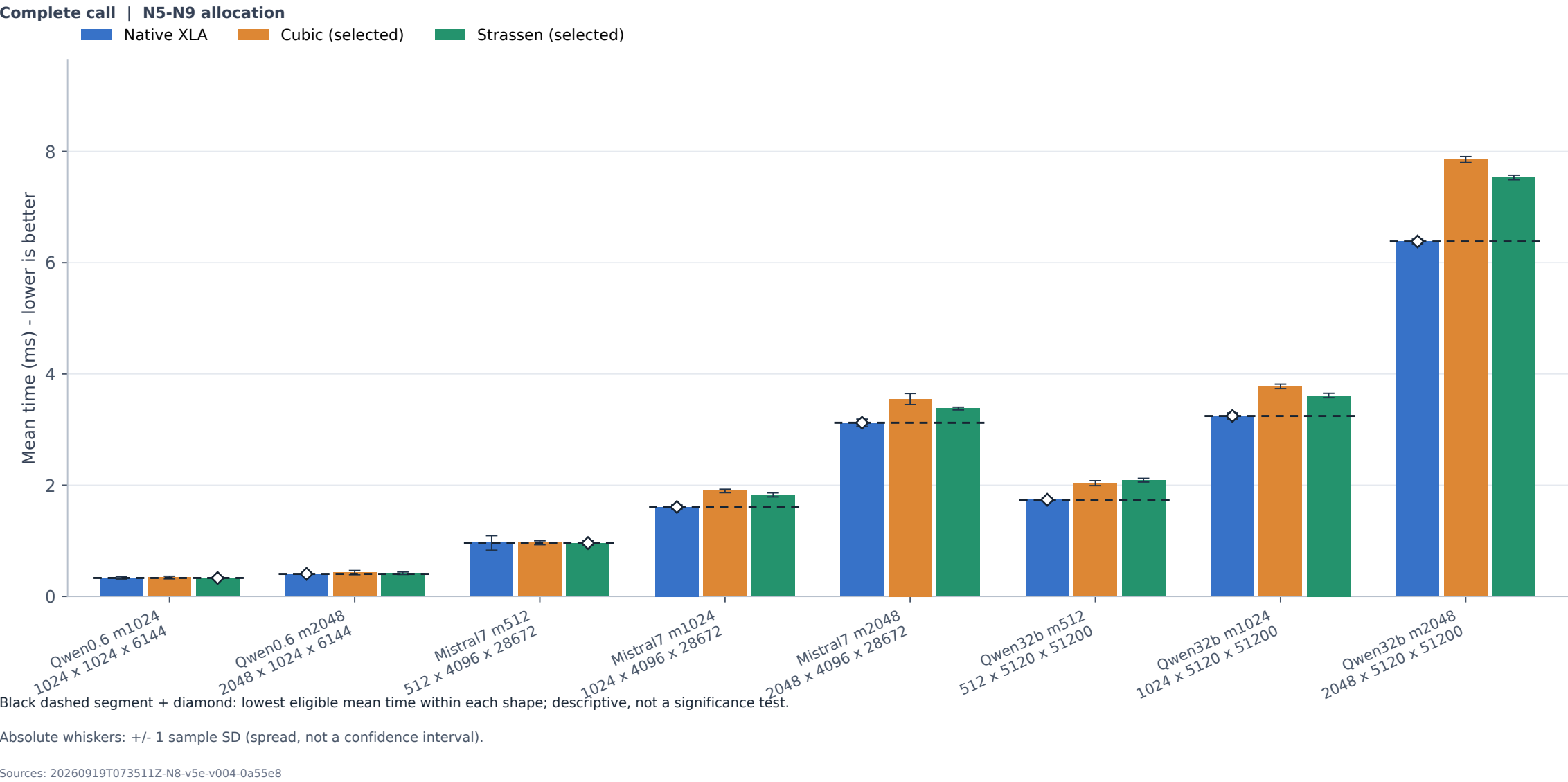
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2, Native XLA 2, Frozen rule 2. Archived paired intervals for the independently tuned Strassen alternative versus Native: 3 faster, 2 slower, 1 inconclusive.

OUR INTERPRETATION

No retuning or rule update occurred on this fresh v5e allocation. The same three custom complete-call gains repeat: deep-K Cubic 1.080x, tall Strassen 1.115x and square Strassen 1.087x versus Native. A nominal extra Native-versus-Native win is timing variability, not a new algorithmic gain. This is fresh-v5e replication; v6e remains future work.

N8 | swiglu - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.



WHAT THIS GRAPH SHOWS

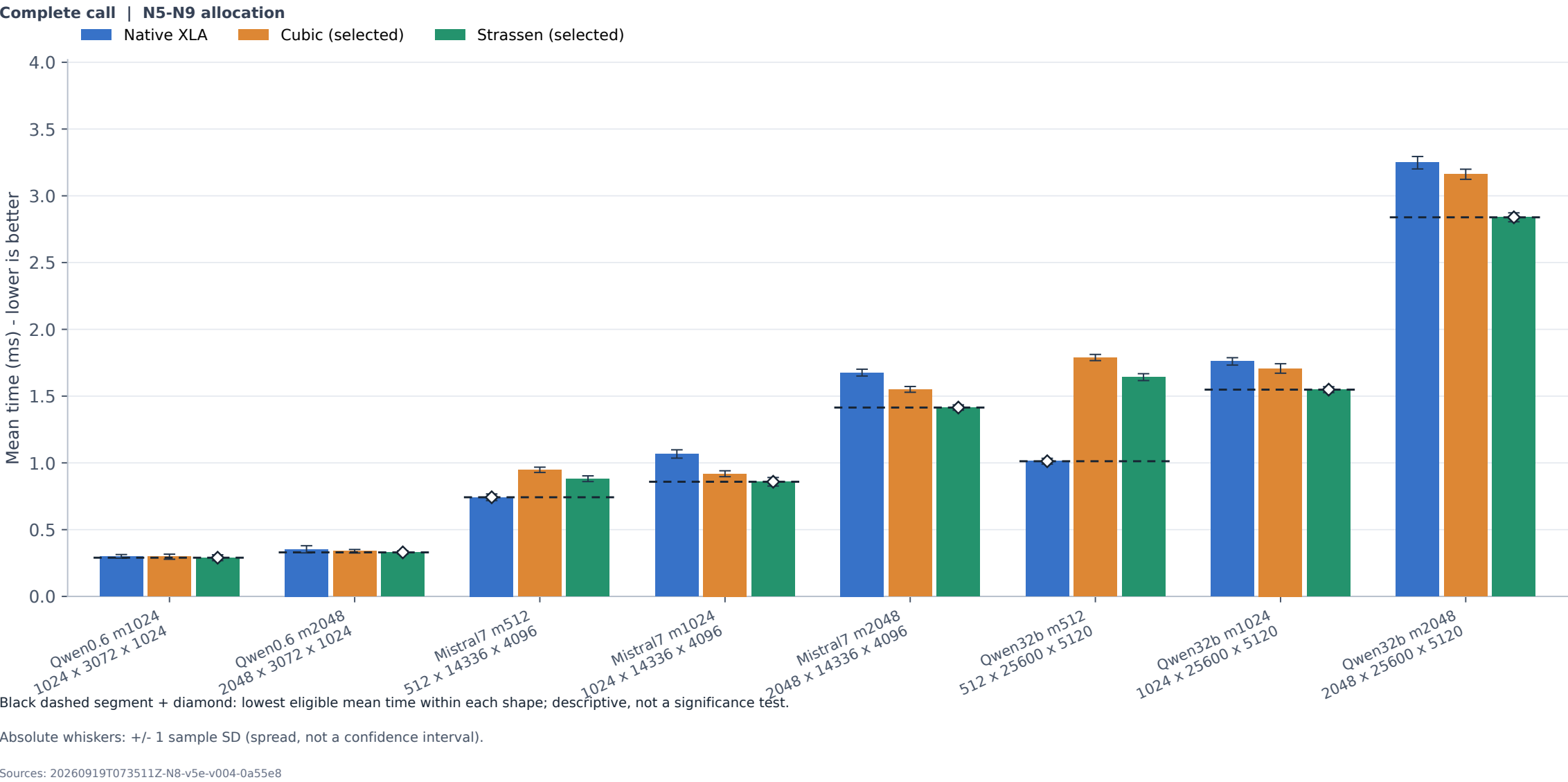
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 2, Native XLA 6. Archived paired intervals for Strassen versus Native: 0 faster, 6 slower, 2 inconclusive.

OUR INTERPRETATION

These bars choose an implementation within each family using the same N8 measurements; they are not fresh confirmation. Separate matched-fusion tests show that Strassen is usually faster than custom Cubic, but Native remains essential: fused Strassen loses all 8 complete-call SwiGLU cases and wins 6 of 8 residual cases. Synthetic activations here do not establish real-model quality.

N8 | residual - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.



WHAT THIS GRAPH SHOWS

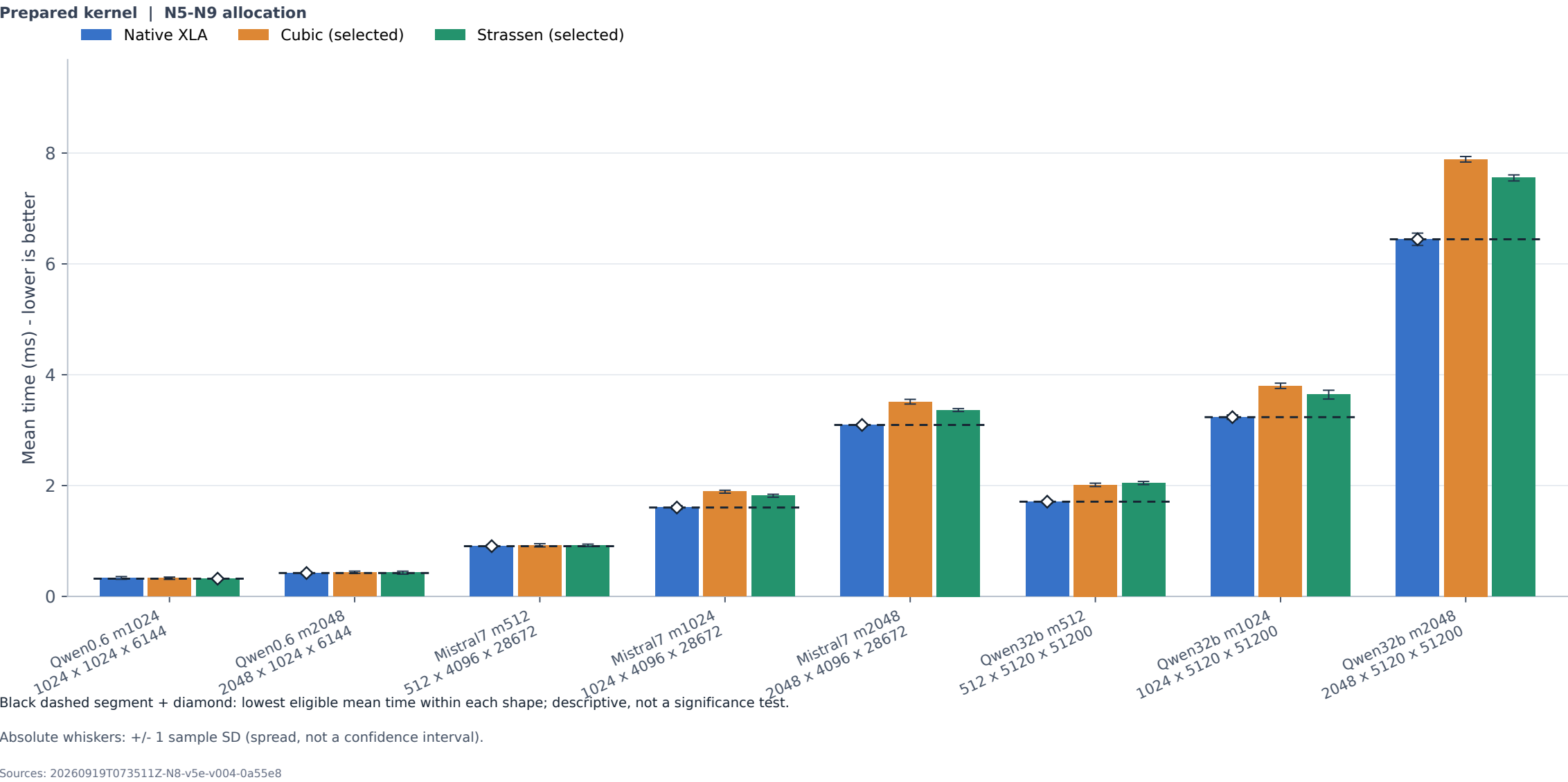
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 6, Native XLA 2. Archived paired intervals for Strassen versus Native: 6 faster, 2 slower, 0 inconclusive.

OUR INTERPRETATION

Residual addition is the more favorable application pattern. In the separate matched-fused comparison, Strassen beats Native on 6 of 8 shapes and loses on the two M=512 cases. The selected-family bars here use the same selection data, and may choose different variants; they should not be mistaken for independent confirmation or an isolated arithmetic effect.

N8 | swiglu - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.



WHAT THIS GRAPH SHOWS

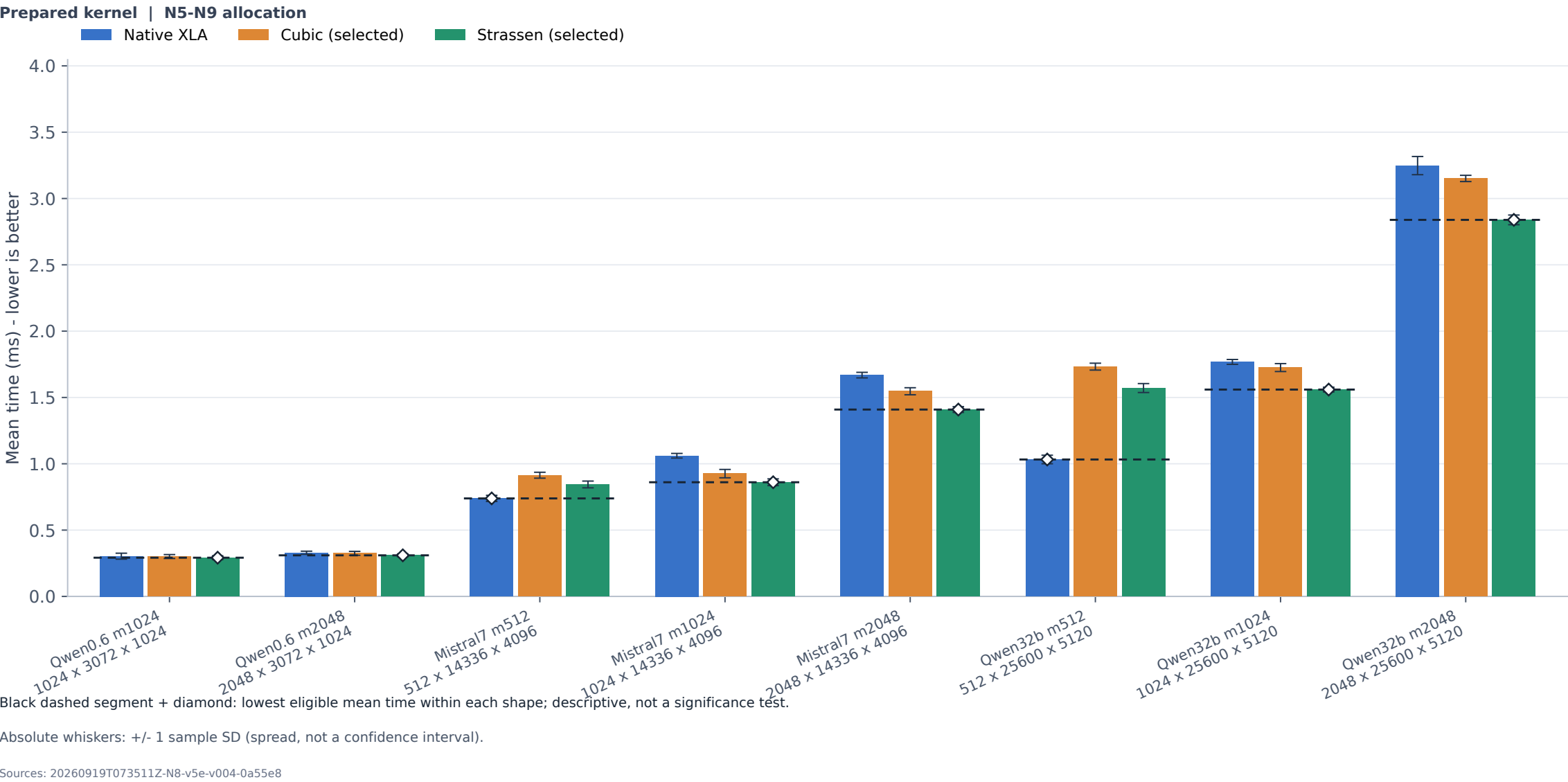
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 1, Native XLA 7. Archived paired intervals for Strassen versus Native: 1 faster, 6 slower, 1 inconclusive.

OUR INTERPRETATION

Reusing prepared operands can materially improve SwiGLU comparisons. The separate matched-fused test gives Strassen 5 wins, 1 inconclusive comparison and 2 losses versus Native in this scope, despite losing all 8 complete-call cases. These selected-family bars retain choices made by complete-call time. A deployment benefit requires an explicit assumption that preparation can be reused.

N8 | residual - selected family implementations

N8-selected on these same measurements; not fresh confirmation. Prepared plots retain complete-call selections.



WHAT THIS GRAPH SHOWS

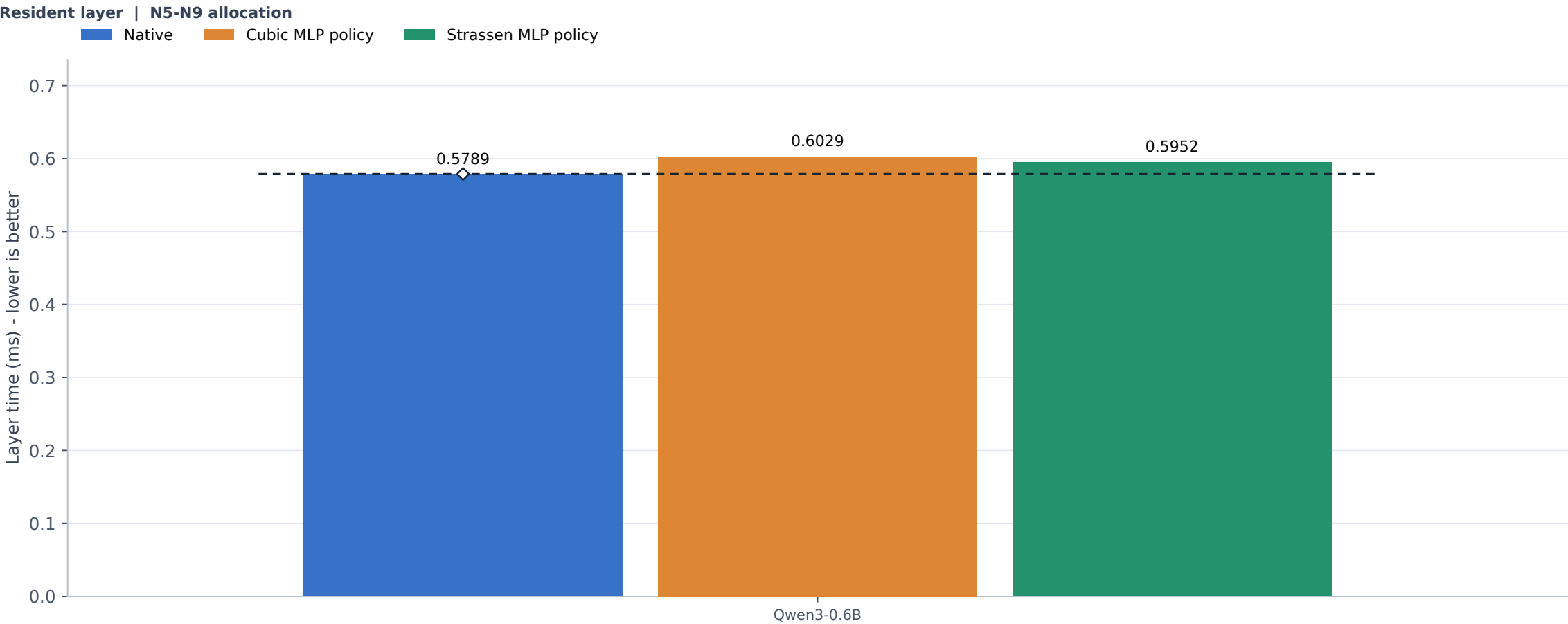
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen (selected) 6, Native XLA 2. Archived paired intervals for Strassen versus Native: 6 faster, 2 slower, 0 inconclusive.

OUR INTERPRETATION

Residual addition is the more favorable application pattern. In the separate matched-fused comparison, Strassen beats Native on 6 of 8 shapes and loses on the two M=512 cases. The selected-family bars here use the same selection data, and may choose different variants; they should not be mistaken for independent confirmation or an isolated arithmetic effect.

N9 | Qwen3-0.6B - resident layer

Complete transformer layer; weights resident. Mean across 28 or 32 different layers; layers are not independent replicates.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

No pooled layer confidence interval is inferred.

Sources: 20260919T081646Z-N9-v5e-v004-0fad6

WHAT THIS GRAPH SHOWS

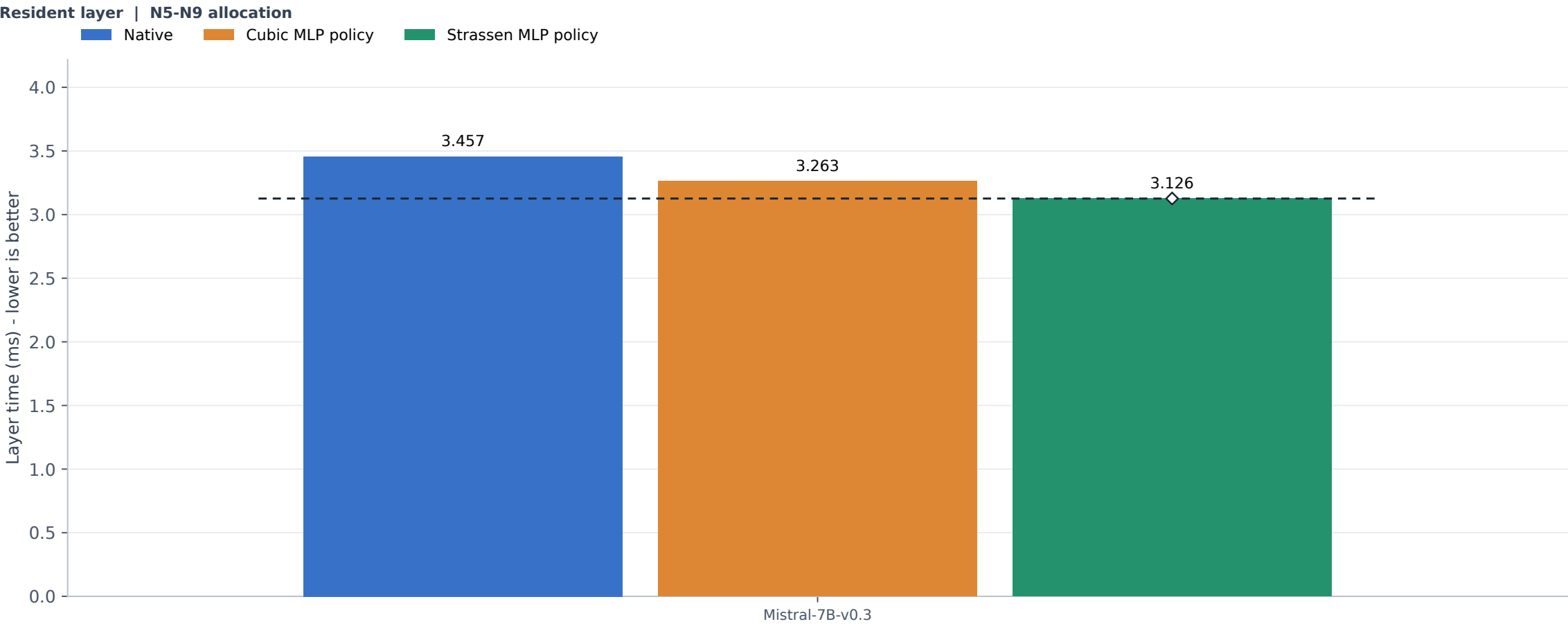
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native 1. Native time / Strassen time = 0.9727x; above 1 favors Strassen. This ratio is descriptive.

OUR INTERPRETATION

Strassen is about 2.81% slower than Native on Qwen, although slightly faster than the tested Cubic policy. This is a mean across different resident transformer layers; there is no pooled confidence interval and no whole-model speedup claim.

N9 | Mistral-7B-v0.3 - resident layer

Complete transformer layer; weights resident. Mean across 28 or 32 different layers; layers are not independent replicates.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

No pooled layer confidence interval is inferred.

Sources: 20260919T081646Z-N9-v5e-v004-0fad6

WHAT THIS GRAPH SHOWS

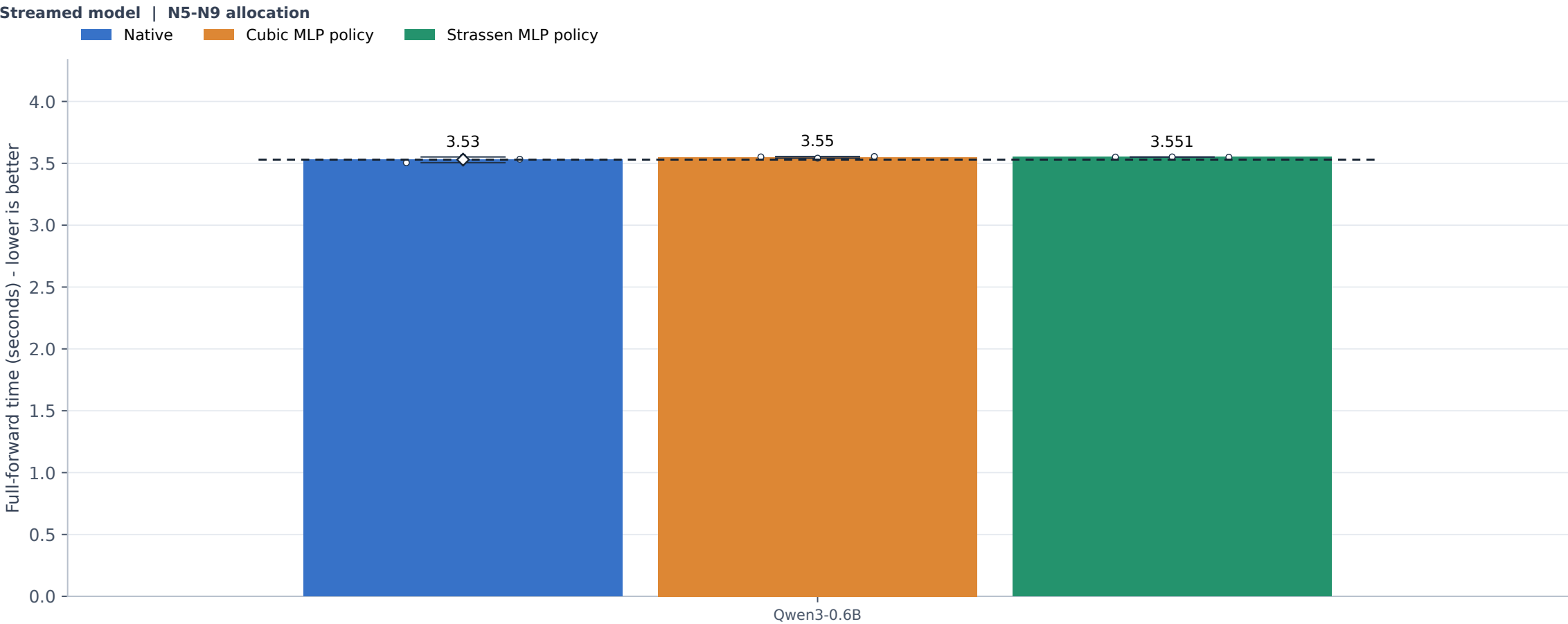
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Strassen MLP policy 1. Native time / Strassen time = 1.1057x; above 1 favors Strassen. This ratio is descriptive.

OUR INTERPRETATION

Strassen gives a descriptive 1.106x speedup over Native and 1.044x over the tested Cubic policy on resident Mistral layers. This is a useful application result, but the layers are different workloads rather than independent replicates. It does not automatically translate into transfer-inclusive model speedup.

N9 | Qwen3-0.6B - streamed full forward

1,024-token full forward including reads, host layout and transfers. All three measured repeats are shown.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Whiskers: minimum to maximum of 3 repeats; descriptive, not a confidence interval.

Sources: 20260919T081646Z-N9-v5e-v004-0fad6

WHAT THIS GRAPH SHOWS

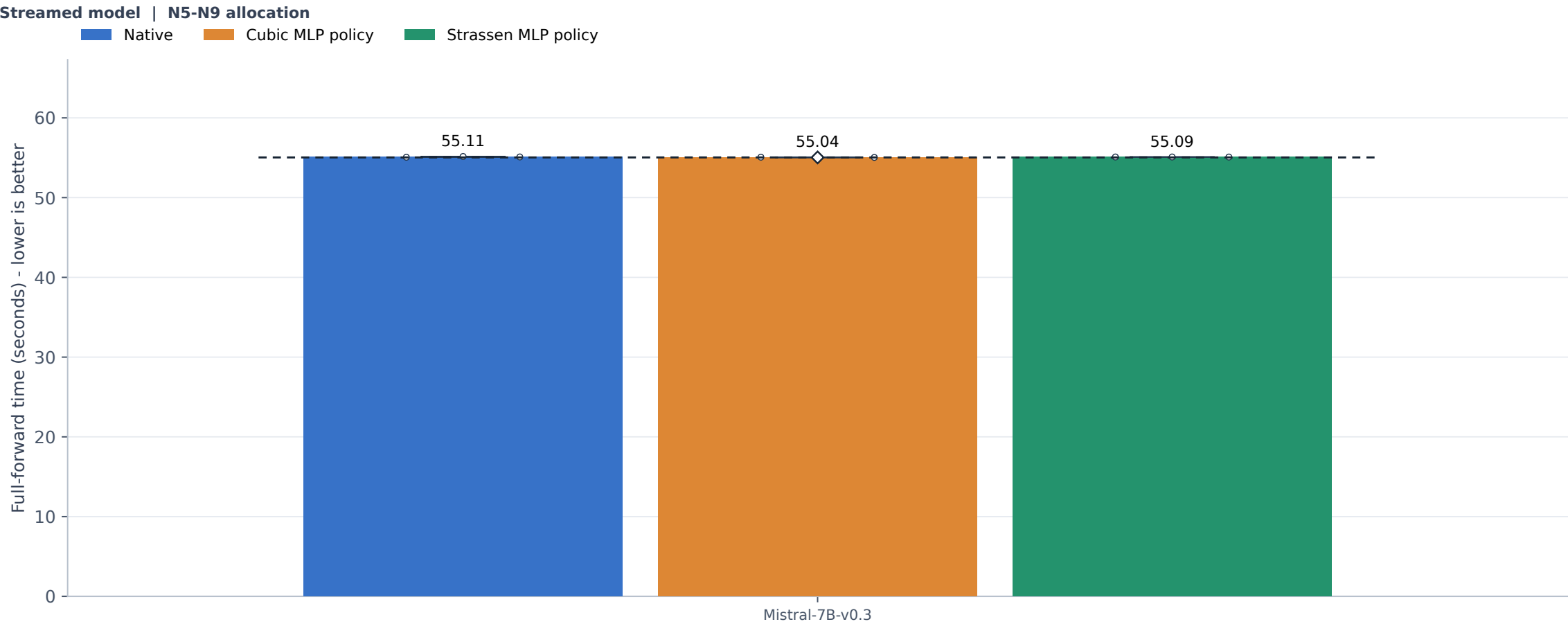
The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Native 1. Native time / Strassen time = 0.9941x; above 1 favors Strassen. This ratio is descriptive.

OUR INTERPRETATION

The three policies are near parity in the transfer-inclusive full forward. Compared with Native, Strassen changes the Qwen mean by +0.591% and the Mistral mean by -0.028%. Only three repeats were recorded per policy. The lowest-mean line makes the descriptive ordering visible; it is not evidence of a reliable serving-speedup gain.

N9 | Mistral-7B-v0.3 - streamed full forward

1,024-token full forward including reads, host layout and transfers. All three measured repeats are shown.



Black dashed segment + diamond: lowest eligible mean time within each shape; descriptive, not a significance test.

Whiskers: minimum to maximum of 3 repeats; descriptive, not a confidence interval.

Sources: 20260919T081646Z-N9-v5e-v004-0fad6

WHAT THIS GRAPH SHOWS

The black segments mark the lowest observed mean within each group. Lowest-mean counts on this page: Cubic MLP policy 1. Native time / Strassen time = 1.0003x; above 1 favors Strassen. This ratio is descriptive.

OUR INTERPRETATION

The three policies are near parity in the transfer-inclusive full forward. Compared with Native, Strassen changes the Qwen mean by +0.591% and the Mistral mean by -0.028%. Only three repeats were recorded per policy. The lowest-mean line makes the descriptive ordering visible; it is not evidence of a reliable serving-speedup gain.

N9 | Corpus quality - nll delta

32,736 scored positions per model and policy. Native zeros are self-comparisons; independent qualification is separate.



Dotted line: predeclared aggregate gate. No statistical interval is inferred.

Sources: 20260919T081646Z-N9-v5e-v004-0fad6

WHAT THIS GRAPH SHOWS

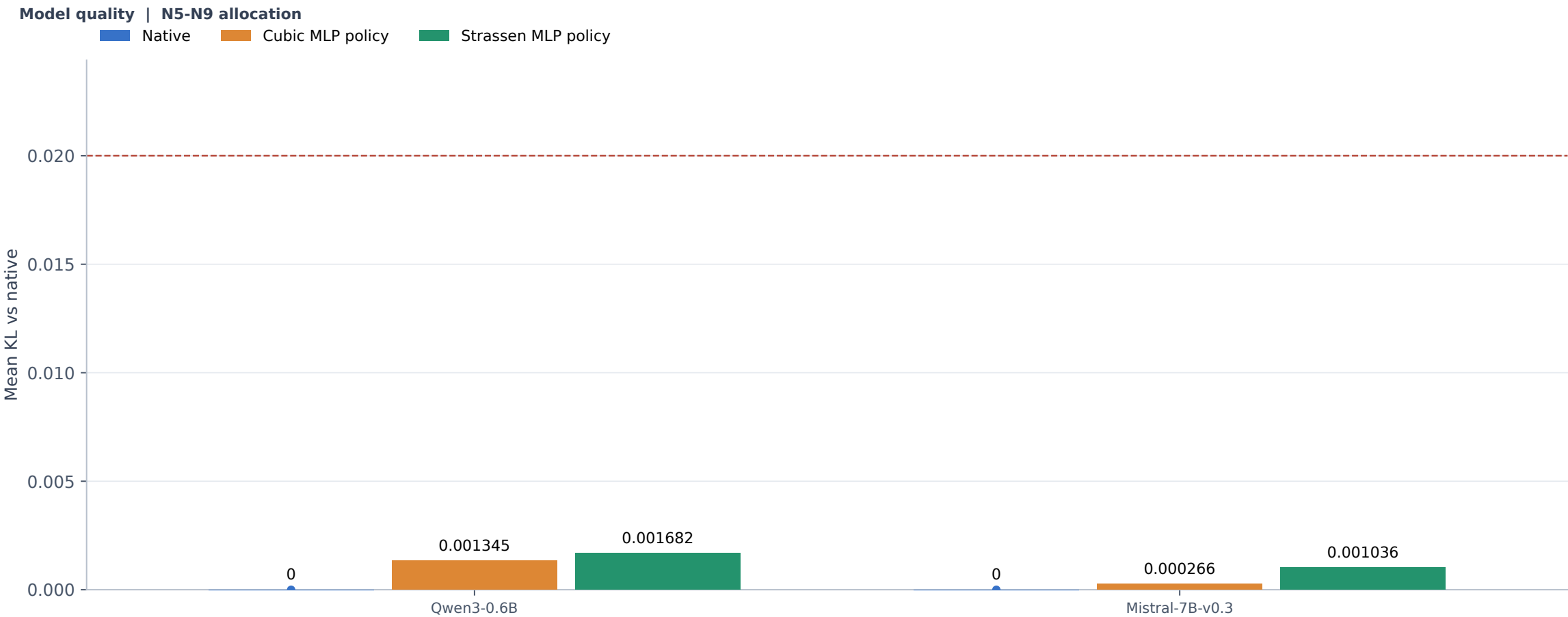
Strassen values are Qwen3-0.6B: 0.00022559; Mistral-7B-v0.3: 0.00029719. The sign of NLL difference matters; the acceptance gate uses its absolute magnitude. Native is zero by self-comparison. Red lines show acceptance gates; these are not timing races.

OUR INTERPRETATION

Both custom policies pass all measured aggregate corpus gates over 32,736 positions per model. Strassen top-1 agreement is 97.73% on Qwen and 98.94% on Mistral, with NLL differences below 0.0003 nats/token. This supports tolerance-based agreement on this corpus, not identical predictions, task accuracy, or universal numerical safety.

N9 | Corpus quality - mean kl

32,736 scored positions per model and policy. Native zeros are self-comparisons; independent qualification is separate.



Dotted line: predeclared aggregate gate. No statistical interval is inferred.

Sources: 20260919T081646Z-N9-v5e-v004-0fad6

WHAT THIS GRAPH SHOWS

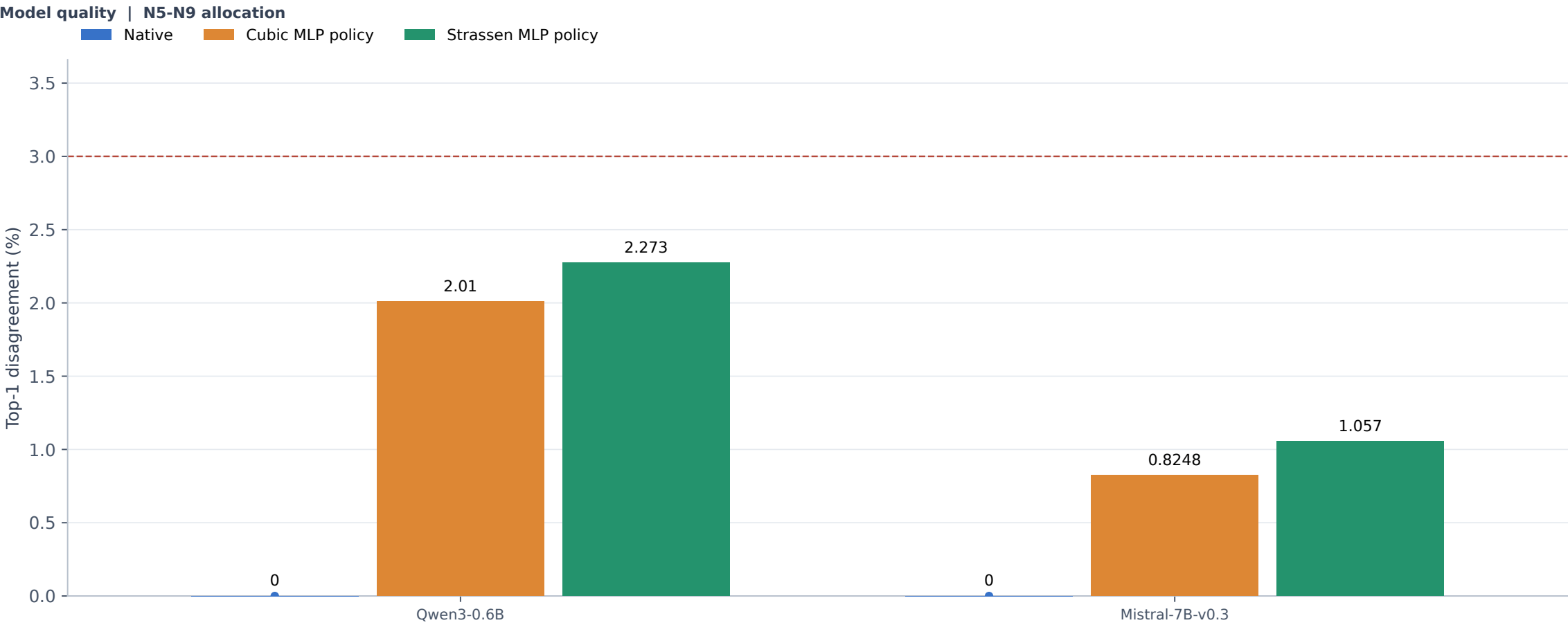
Strassen values are Qwen3-0.6B: 0.0016817; Mistral-7B-v0.3: 0.0010361. Lower error or disagreement means closer agreement with Native. Native is zero by self-comparison. Red lines show acceptance gates; these are not timing races.

OUR INTERPRETATION

Both custom policies pass all measured aggregate corpus gates over 32,736 positions per model. Strassen top-1 agreement is 97.73% on Qwen and 98.94% on Mistral, with NLL differences below 0.0003 nats/token. This supports tolerance-based agreement on this corpus, not identical predictions, task accuracy, or universal numerical safety.

N9 | Corpus quality - top1 disagreement

32,736 scored positions per model and policy. Native zeros are self-comparisons; independent qualification is separate.



WHAT THIS GRAPH SHOWS

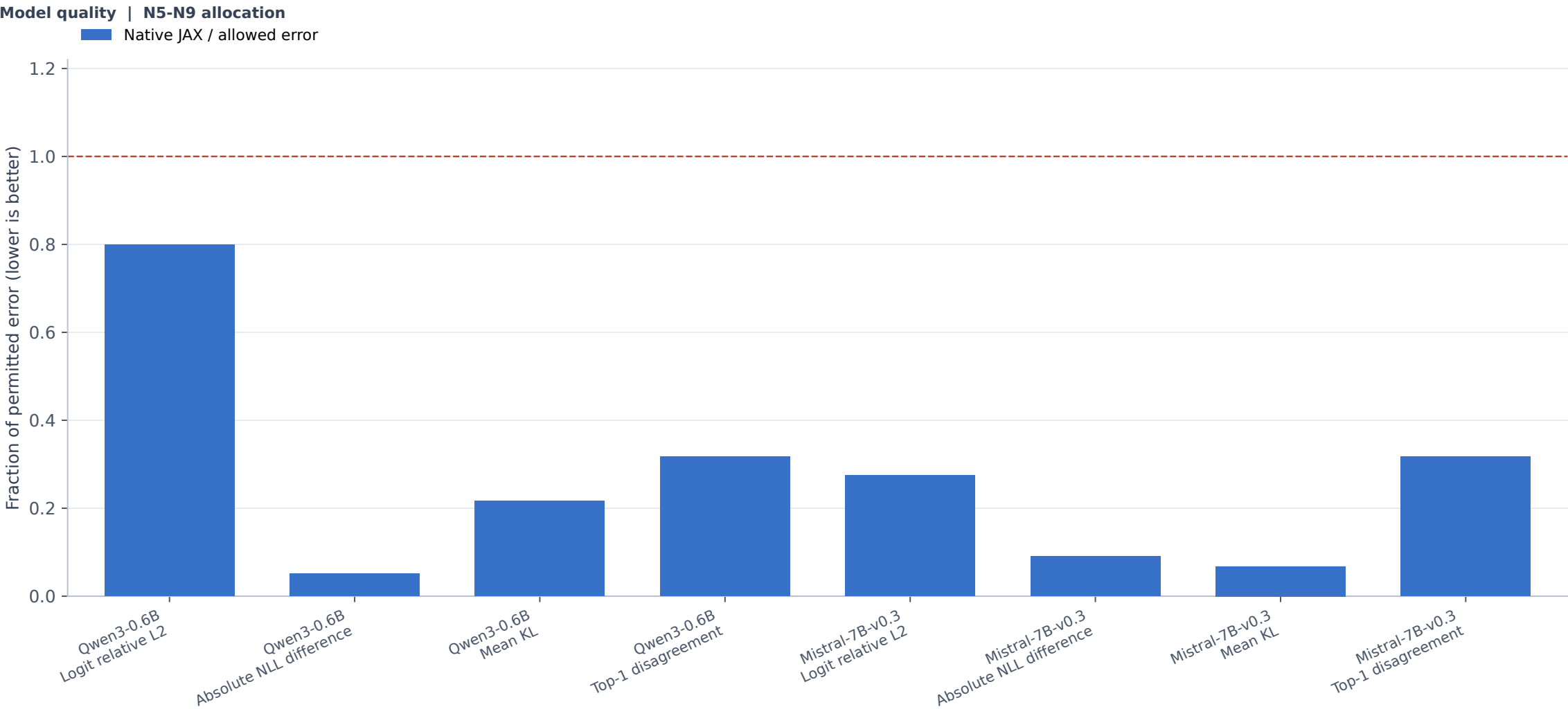
Strassen values are Qwen3-0.6B: 2.2727; Mistral-7B-v0.3: 1.0569. Lower error or disagreement means closer agreement with Native. Native is zero by self-comparison. Red lines show acceptance gates; these are not timing races.

OUR INTERPRETATION

Both custom policies pass all measured aggregate corpus gates over 32,736 positions per model. Strassen top-1 agreement is 97.73% on Qwen and 98.94% on Mistral, with NLL differences below 0.0003 nats/token. This supports tolerance-based agreement on this corpus, not identical predictions, task accuracy, or universal numerical safety.

N9 | Independent native implementation qualification

Native JAX vs official Transformers CPU BF16 on 64 input tokens / 63 scored positions.



The limits differ from the full-corpus custom-policy gates. Finite-output checks also passed.

Sources: 20260919T081646Z-N9-v5e-v004-0fad6

WHAT THIS GRAPH SHOWS

Each bar is one Native JAX error divided by its allowed limit, checked against official Transformers CPU BF16. Every bar is below 1, so both native implementations pass this short 63-position qualification. These are different checks and tolerances from the full-corpus policy comparison.

OUR INTERPRETATION

This independent check gives a basis for using the Native implementation as a comparison reference. It covers only the first 64 input tokens (63 scored positions), not the complete corpus. The full-corpus custom-policy checks and N4 cancellation tests answer separate questions.